

Chapter I

Introduction

A note on the use of these ppt slides:

We're making these slides freely available to all (faculty, students, readers). They're in PowerPoint form so you see the animations; and can add, modify, and delete slides (including this one) and slide content to suit your needs. They obviously represent a *lot* of work on our part. In return for use, we only ask the following:

- ❖ If you use these slides (e.g., in a class) that you mention their source (after all, we'd like people to use our book!)
- ❖ If you post any slides on a www site, that you note that they are adapted from (or perhaps identical to) our slides, and note our copyright of this material.

Thanks and enjoy! JFK/KWR

©All material copyright 1996-2012
J.F Kurose and K.W. Ross, All Rights Reserved

Modified by Man Hon Cheung

Chapter 1: introduction

our goal:

- ❖ get “feel” and terminology
- ❖ more depth, detail later in course

overview:

- ❖ what's the Internet?
- ❖ what's a protocol?
- ❖ network edge; hosts, access net, physical media
- ❖ network core: packet/circuit switching, Internet structure
- ❖ performance: loss, delay, throughput
- ❖ security
- ❖ protocol layers, service models
- ❖ history

Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

What's the Internet: "nuts and bolts" view

View 1: Hardware

① **network edge**

❖ millions of connected computing devices:

- **hosts = end systems**
- running **network apps**



PC
server
wireless laptop
smartphone

② **communication links**

❖ **communication links**

- fiber, copper, radio, satellite



wireless links
wired links

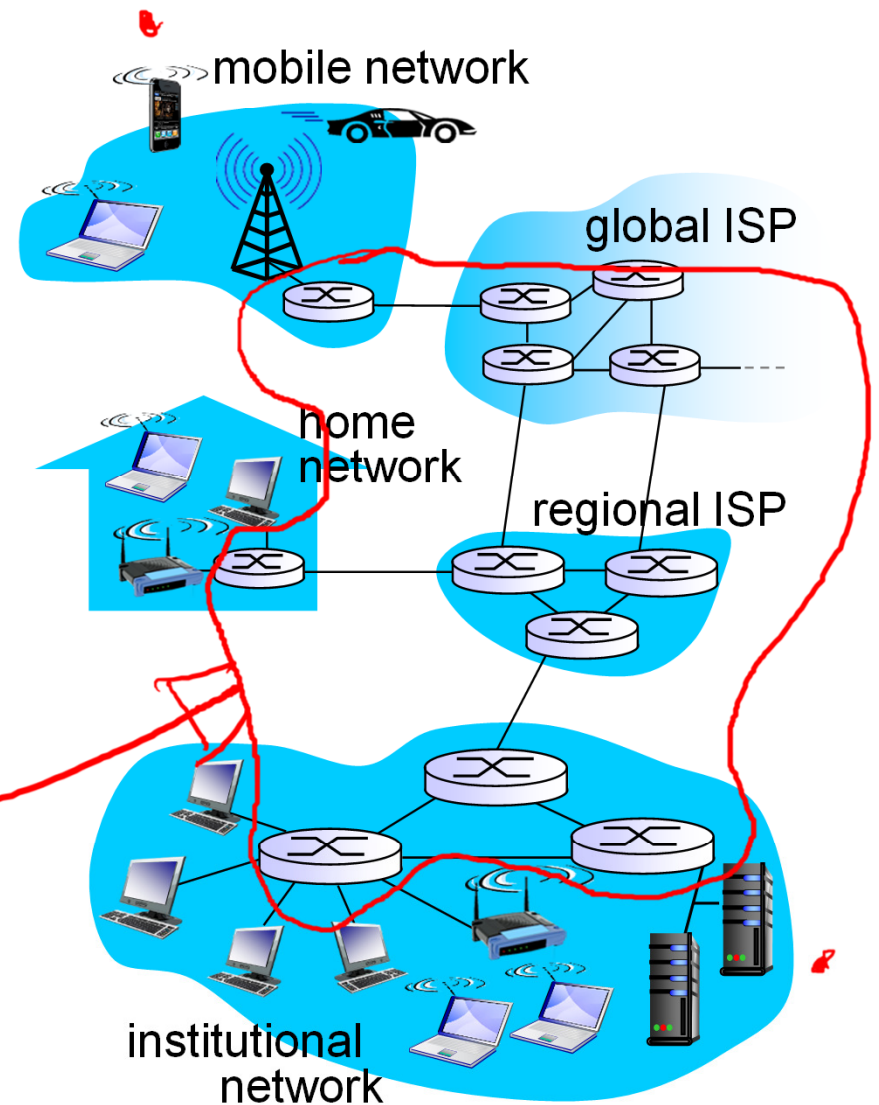
③ **network core**

❖ **Packet switches**: forward packets (chunks of data)

- **routers** and **switches**



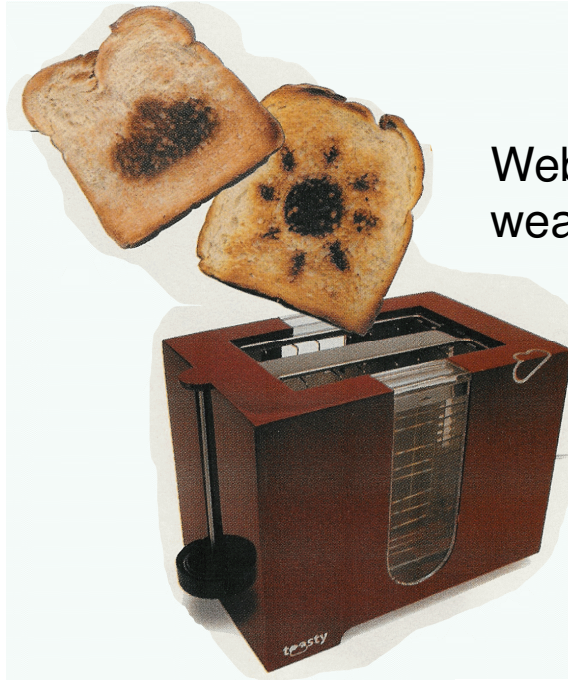
router



“Fun” internet appliances



IP picture frame
<http://www.ceiva.com/>



Web-enabled toaster +
weather forecaster



Tweet-a-watt:
monitor energy use



Internet
refrigerator



Slingbox: watch,
control cable TV remotely



Internet phones

What's the Internet: “nuts and bolts” view

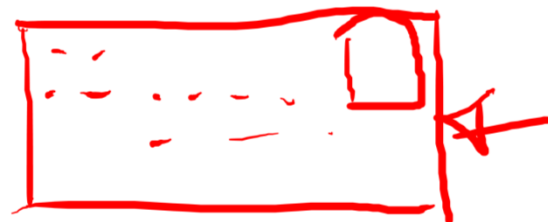
❖ *Internet standards*

- Documents are called **request for comments (RFC)**
- Developed by **Internet Engineering Task Force (IETF)**

What's the Internet: a service view

- ❖ Infrastructure that provides services to applications:
 - Web, VoIP, email, games, e-commerce, social nets, ...
- ❖ Q: How does one program running on one end system instruct the Internet to deliver data to another program running on another end system?
- ❖ A: Internet application programming interface (API): a set of rules that the sending program must follow so that the Internet can deliver the data to the destination

- Analogous to postal service API: Write address, put a stamp,...



socket programming ← TCP (reliable)
UDP (unreliable)

What's a protocol?

human protocols:

- ❖ “what's the time?”
- ❖ “I have a question”
- ❖ introductions

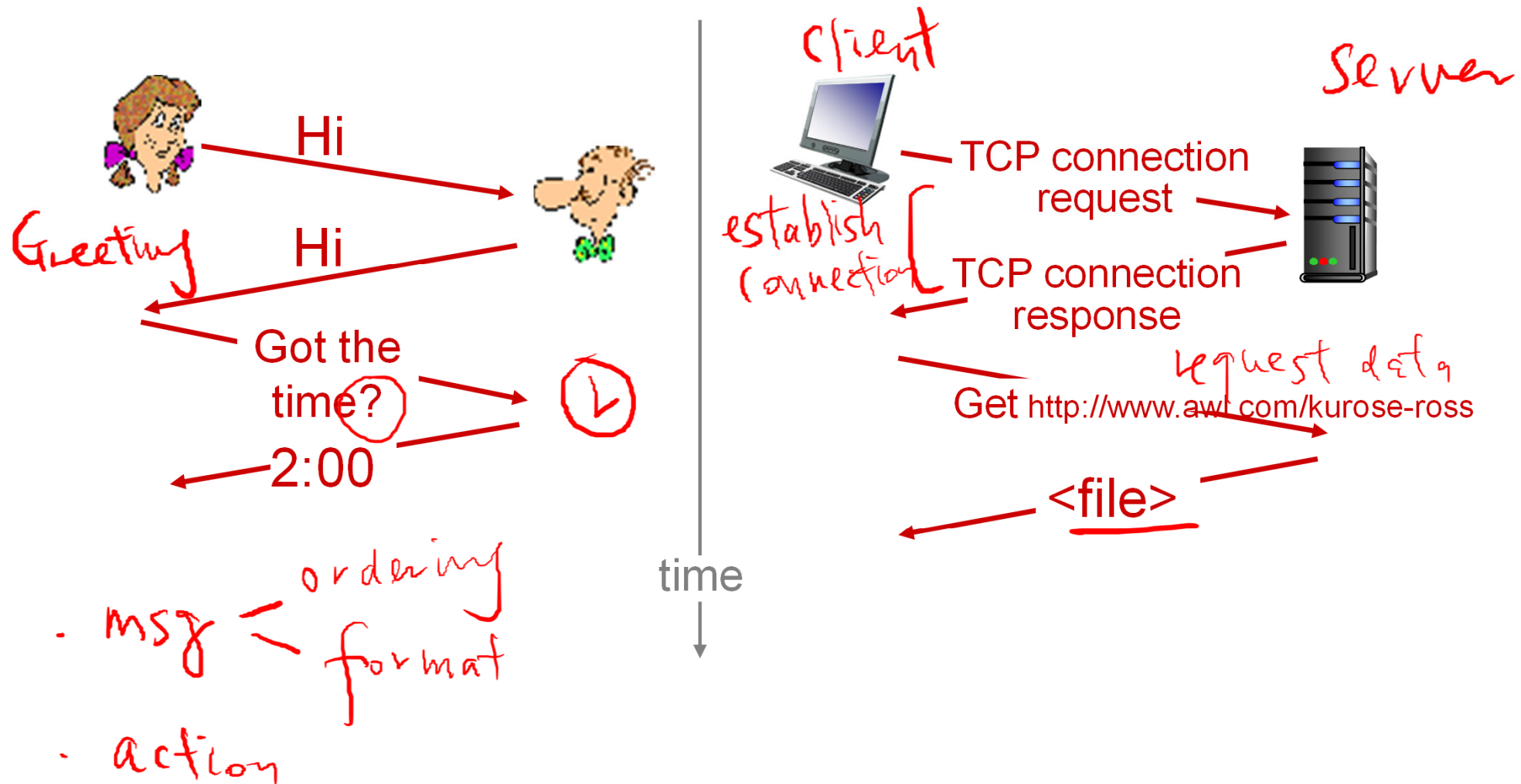
network protocols:

- ❖ machines rather than humans
- ❖ all communication activity in Internet governed by protocols

Protocols define format and order of ^①messages sent and received among network entities, and ^②actions taken on message transmission and receipt.

What's a protocol?

a human protocol and a computer network protocol:



Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

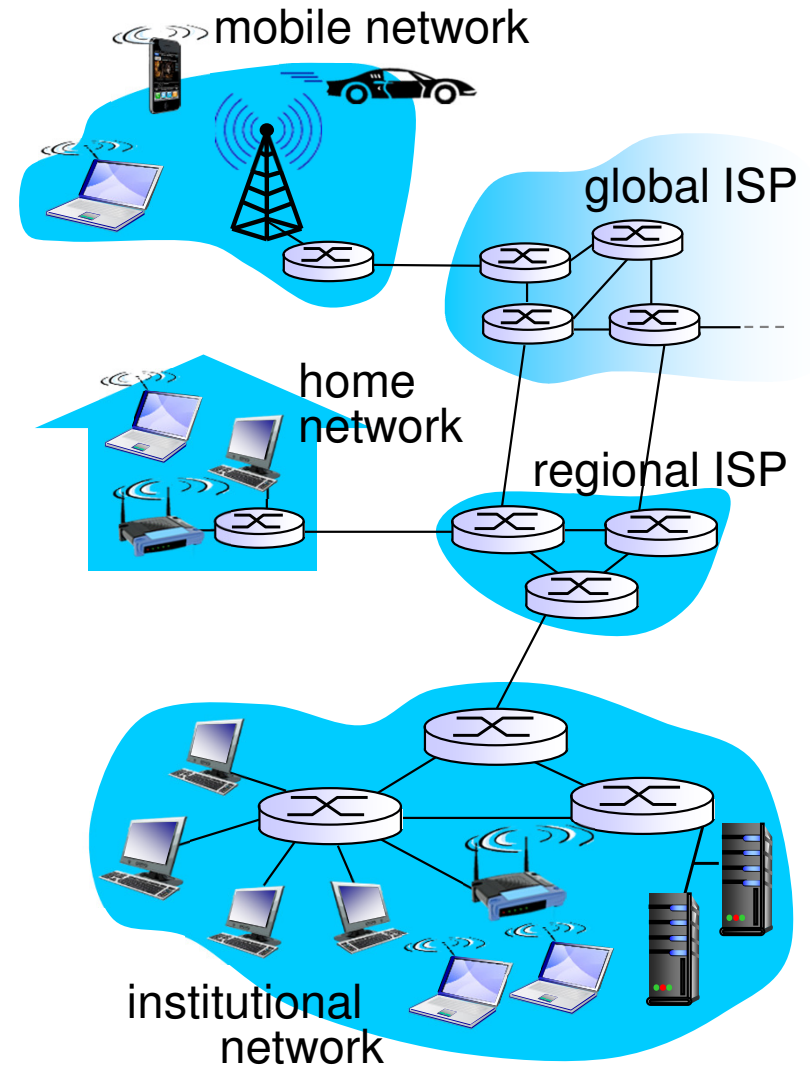
A closer look at network structure:

❖ *network edge:*

- hosts: clients and servers
 - servers often in data centers

❖ *network core:*

- interconnected routers
- network of networks



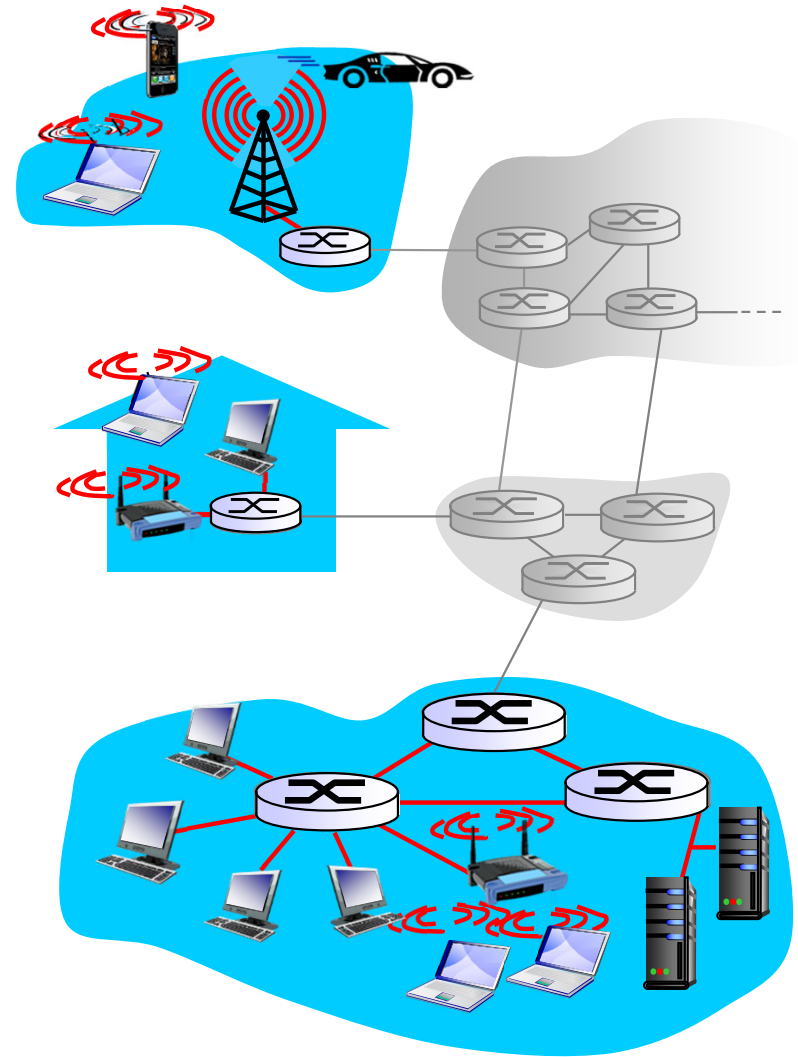
Access networks and physical media

Different types of access networks:

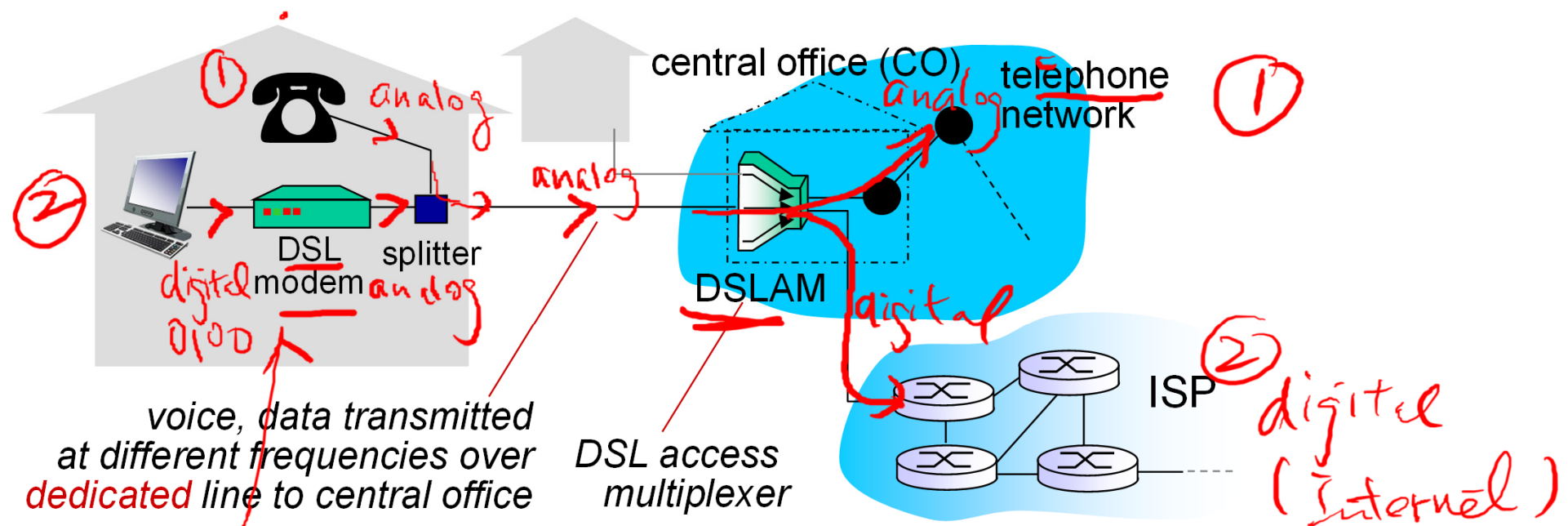
- ❖ residential access networks
- ❖ institutional access networks (school, company)
- ❖ mobile access networks

keep in mind:

- ❖ bandwidth (bits per second) of access network?
- ❖ shared or dedicated?

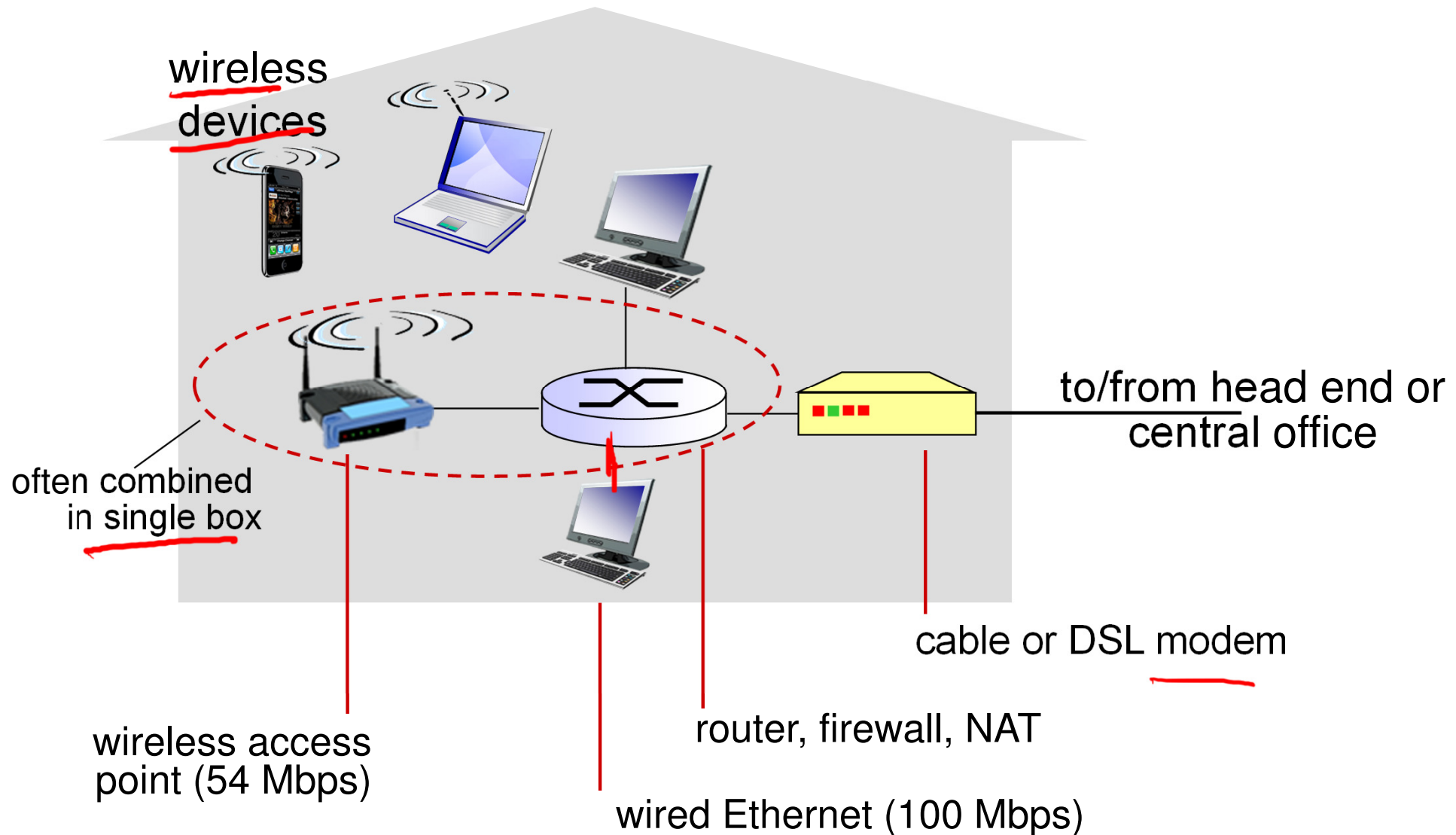


Home access net: digital subscriber line (DSL)

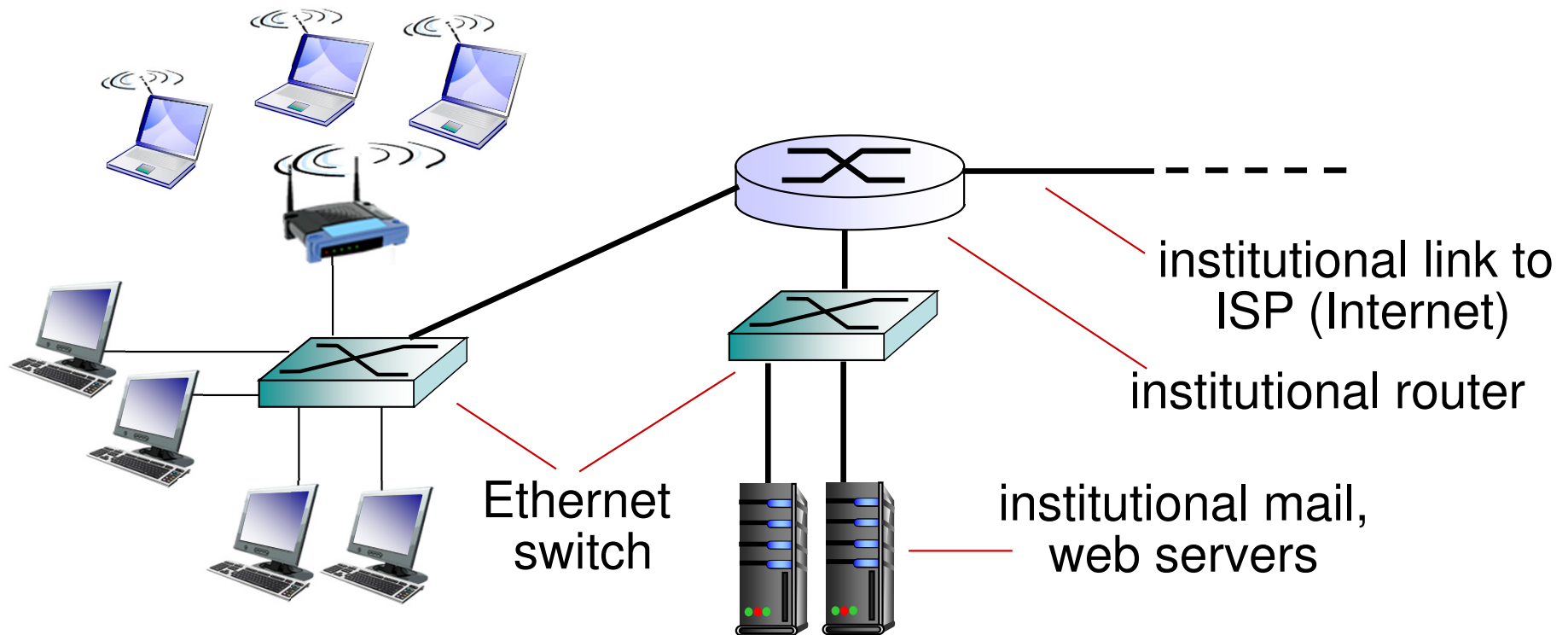


- ❖ DSL modem: takes digital data & translates it to high frequency tones for transmission over telephone wires.
- ❖ use existing telephone line by frequency division multiplexing
 - voice over DSL phone line goes to telephone net
 - data over DSL phone line goes to Internet
- ❖ DSL access multiplexer (DSLAM): Translate analog signals from many such houses into digital format.

Access net: home network



Enterprise access networks (Ethernet)



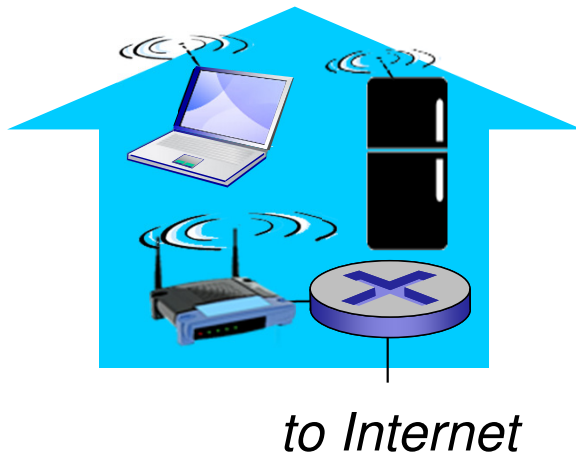
- ❖ **Ethernet** is by far the most prevalent access technology in corporate, university, and home networks
- ❖ 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- ❖ today, end systems typically connect into **Ethernet switch**

Wireless access networks

- ❖ shared *wireless* access network connects end system to router
 - via base station aka “access point”

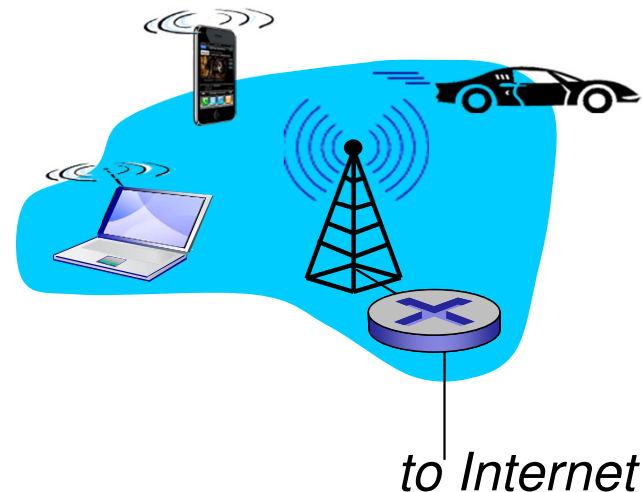
wireless LANs: *Indoor*

- within building (100 ft.)
- 802.11b/g/n/ac (Wi-Fi): 11, 54, 450 Mbps, and around 1 Gbps transmission rate



wide-area wireless access *outdoor*

- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE, *5G*



Physical media

- ❖ **physical link:** what lies between transmitter & receiver
- ❖ Two categories of physical media:
 1. **guided media:**
 - signals propagate in solid media: copper, fiber, coax
 2. **unguided media:**
 - signals propagate freely, e.g., radio

twisted pair (TP)

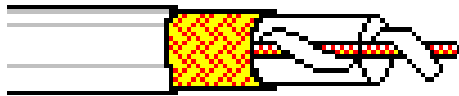
- ❖ least expensive and most commonly used guided transmission medium
- ❖ two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps



Physical media: coax, fiber

coaxial cable:

- ❖ two concentric copper conductors => achieve high data transmission rates
- ❖ bidirectional
- ❖ broadband:
 - multiple channels on cable
 - Hybrid fiber coax (HFC)



fiber optic cable:

- ❖ glass fiber carrying light pulses, each pulse a bit
- ❖ high-speed operation:
 - high-speed point-to-point transmission (e.g., 10's-100's Gbps transmission rate)
- ❖ low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Physical media: radio

- ❖ signal carried in electromagnetic spectrum
- ❖ no physical “wire”
- ❖ bidirectional
- ❖ propagation environment effects:
 - reflection
 - obstruction by objects
 - interference

radio link types:

- LAN (e.g., Wi-Fi)
 - 54 Mbps
- wide-area (e.g., cellular)
 - 4G cellular: ~ 10 Mbps
- satellite
 - Kbps to 45Mbps channel (or multiple smaller channels)
 - 270 msec end-end delay
 - geosynchronous versus low altitude

Chapter 1: roadmap

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching vs circuit switching
- network structure

1.4 delay, loss, throughput in networks

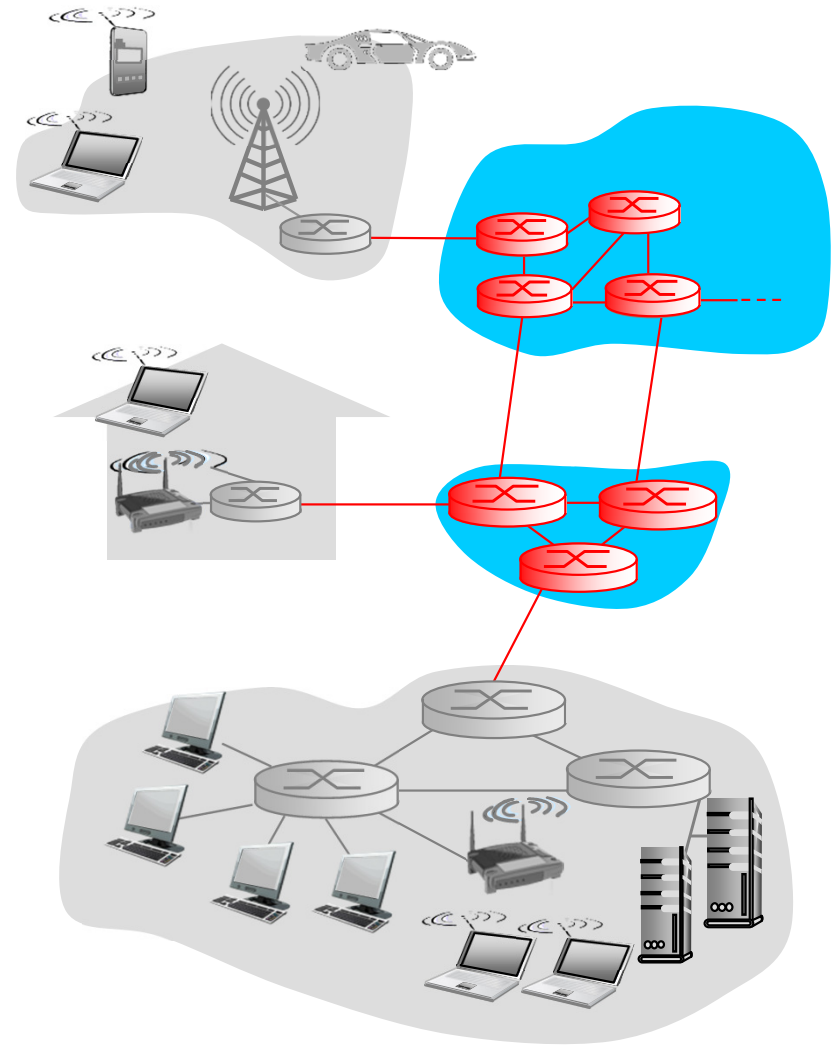
1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

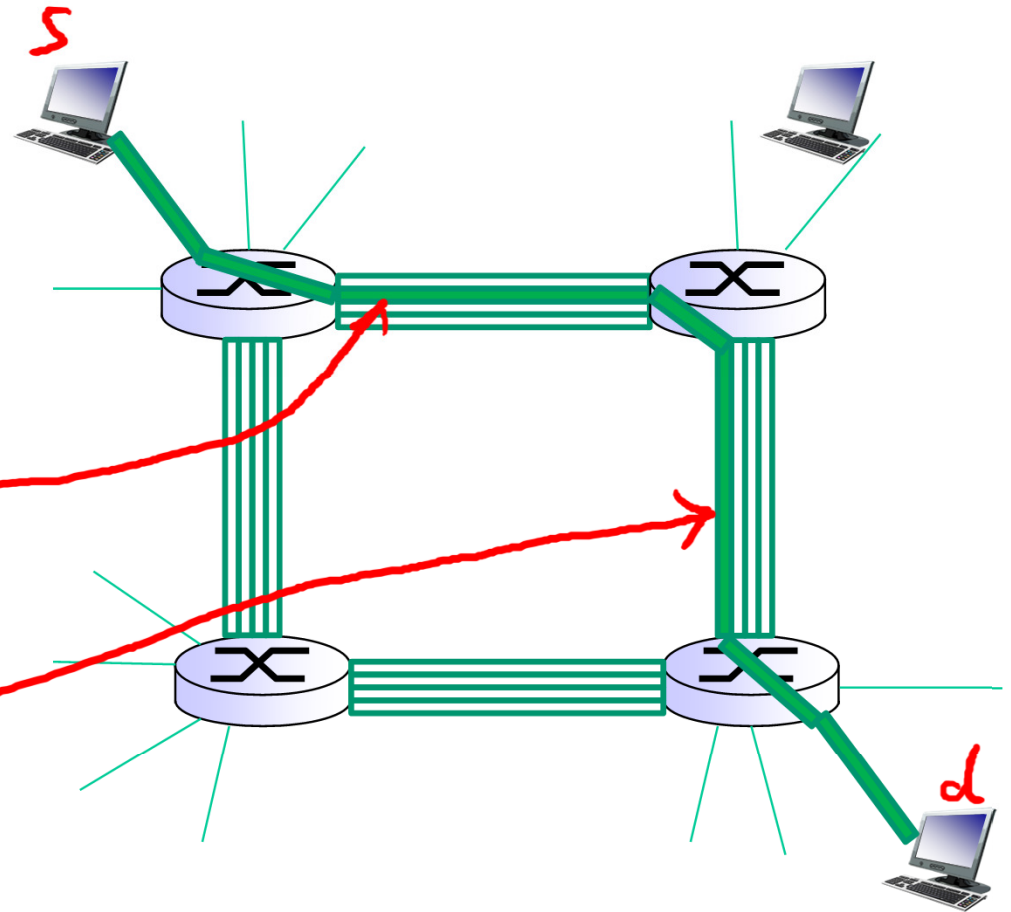
The network core

- ❖ Two fundamental approaches to moving data through a network:
★ packet switching ^{Internet} and circuit switching ^{telephone}.



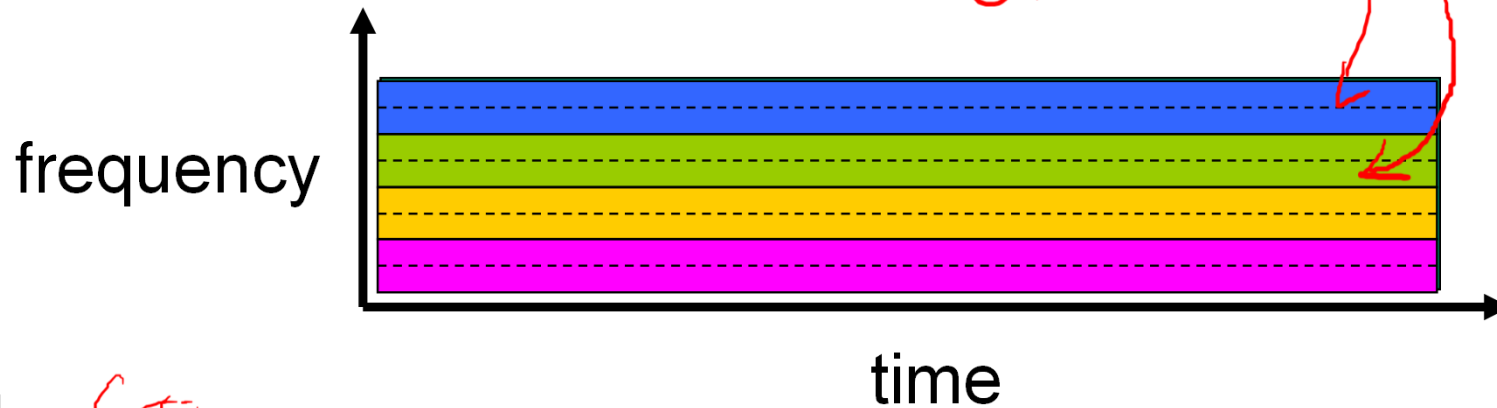
Circuit switching

- ❖ **Circuit switching**: end-end resources are reserved for communication between source & dest:
- ❖ In diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- ❖ circuit segment idle if not used by call (no sharing)
- ❖ Commonly used in traditional telephone networks

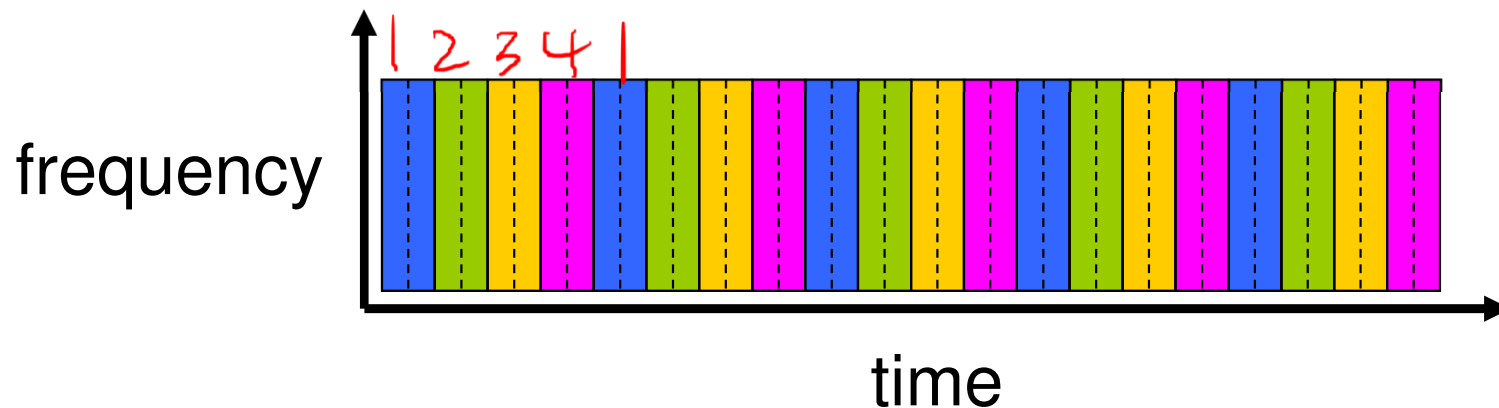


Circuit switching: FDM versus TDM

FDM *(frequency division multiplexing)* Example: 4 users

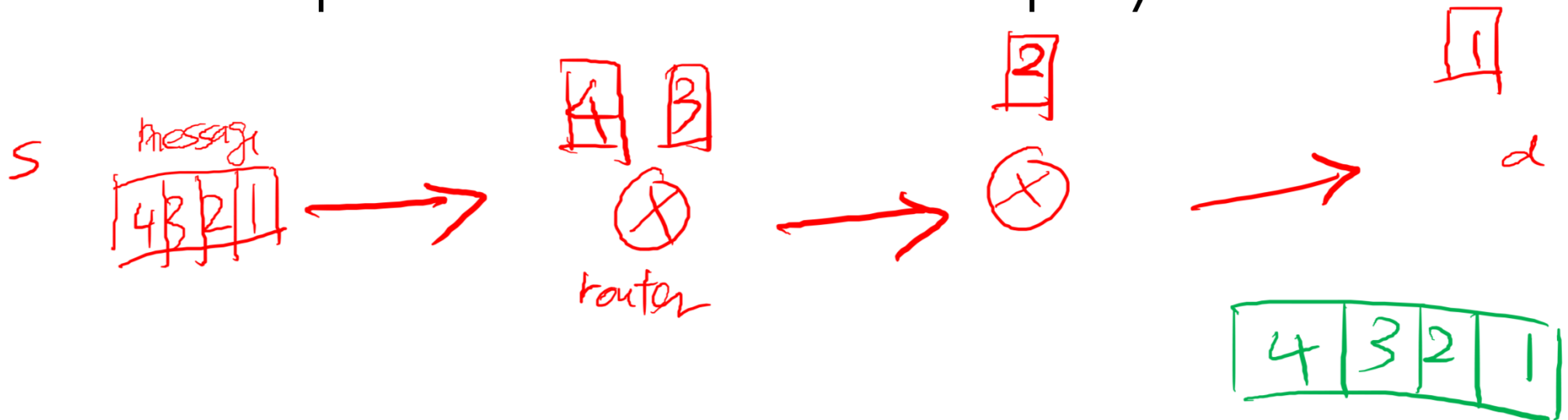


TDM *(time division multiplexing)*



Packet-switching

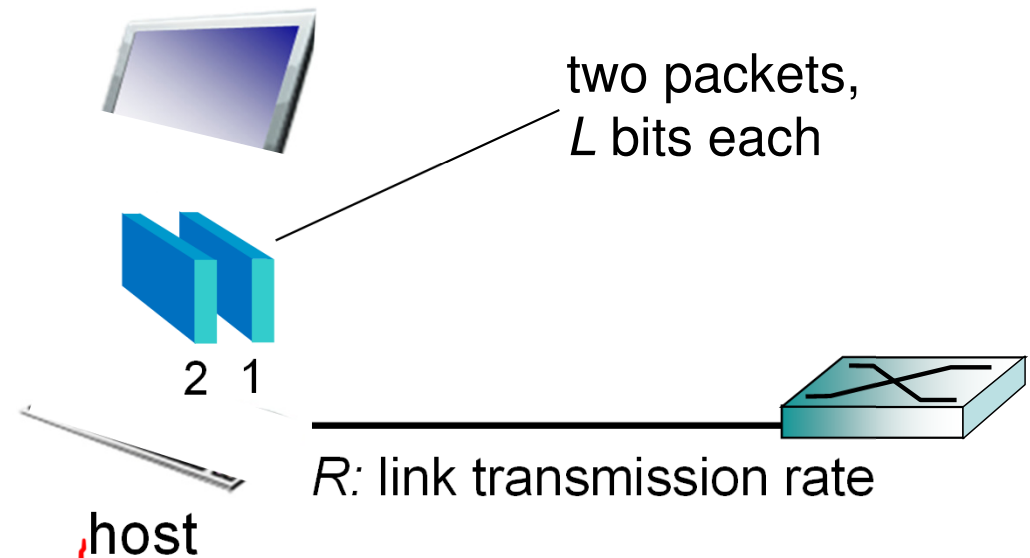
- ❖ **Packet-switching:** hosts break application-layer messages into packets
 - forward packets from one router to the next, across links on path from source to destination
 - each packet transmitted at full link capacity



Host: sends *packets* of data

host sending function:

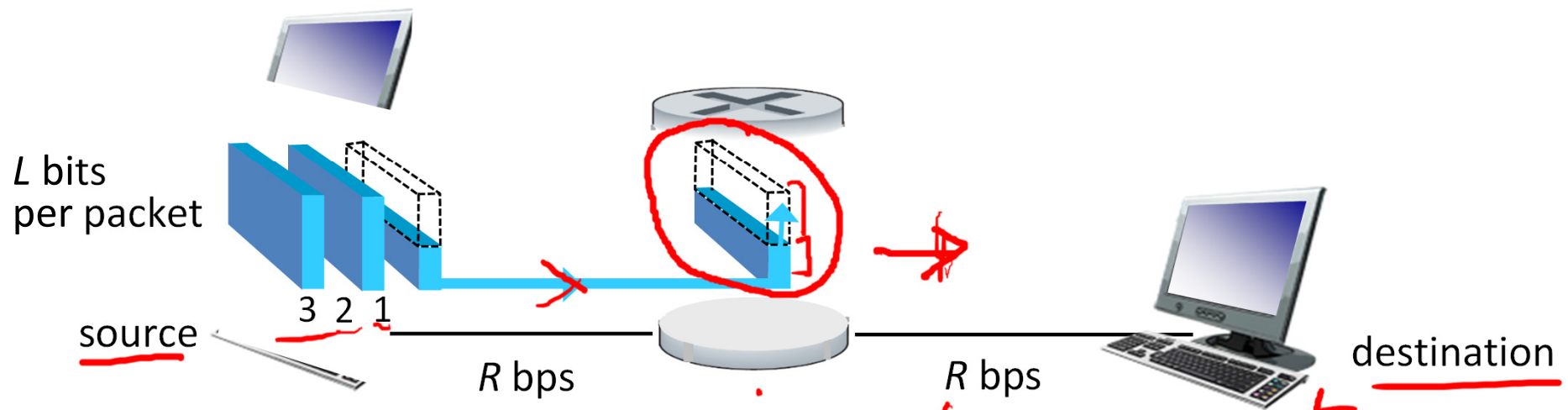
- ❖ takes application message
- ❖ breaks into smaller chunks, known as *packets*, of length L bits $= 1000 \text{ bits}$
- ❖ transmits packet into access network at *transmission rate* $R = 1 \text{ Mbps}$
 - link transmission rate, aka link *capacity*, aka $= 1 \times 10^6 \text{ bits per second}$
link bandwidth



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}} \text{ sec}$$

$$= \frac{1000 \text{ bits}}{1 \times 10^6 \text{ bits/sec}} = 10^{-3} \text{ sec} = 1 \text{ msec}$$

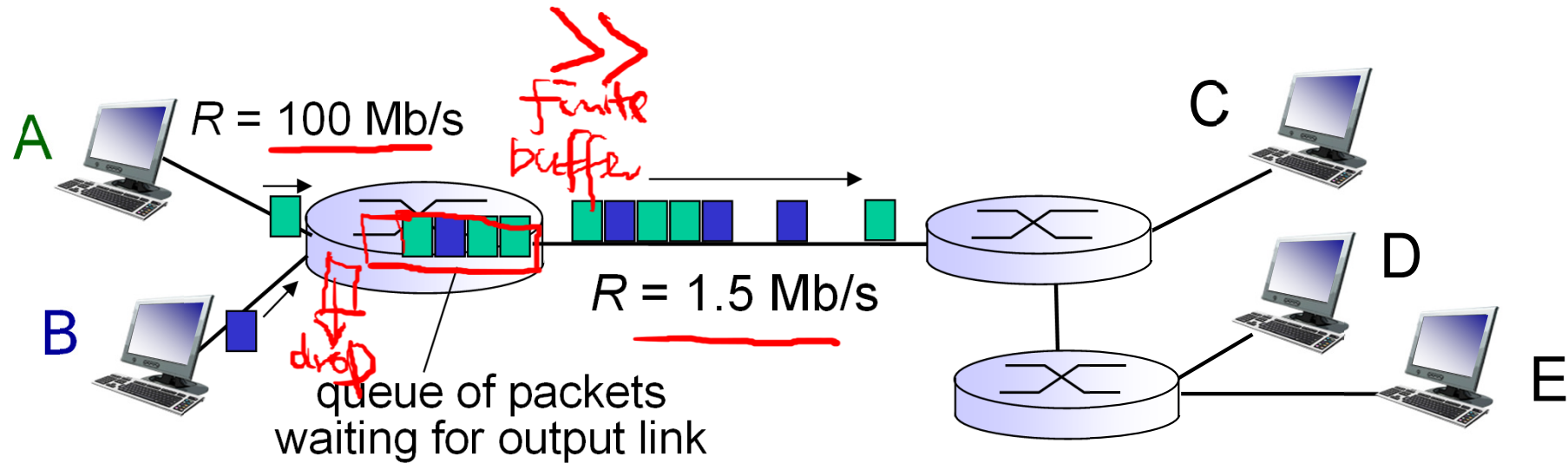
Packet-switching: "store-and-forward"



- ❖ takes L/R seconds to transmit (push out) L -bit packet into link at R bps $t=1$
- ❖ store and forward: entire packet must arrive at router before it can be transmitted on next link $t=2$
- ❖ end-end delay = $\frac{L}{R} + \frac{L}{R}$ (assuming zero propagation delay) $\left. \begin{array}{l} \text{one packet} \end{array} \right\}$ more on delay shortly ...



Packet Switching: queueing delay, loss



- ❖ If arrival rate (in bits) to link exceeds transmission rate of link for a period of time:
 - queuing delay: packets will queue, wait to be transmitted on link
 - Packet loss: packets can be dropped (lost) if memory (buffer) fills up

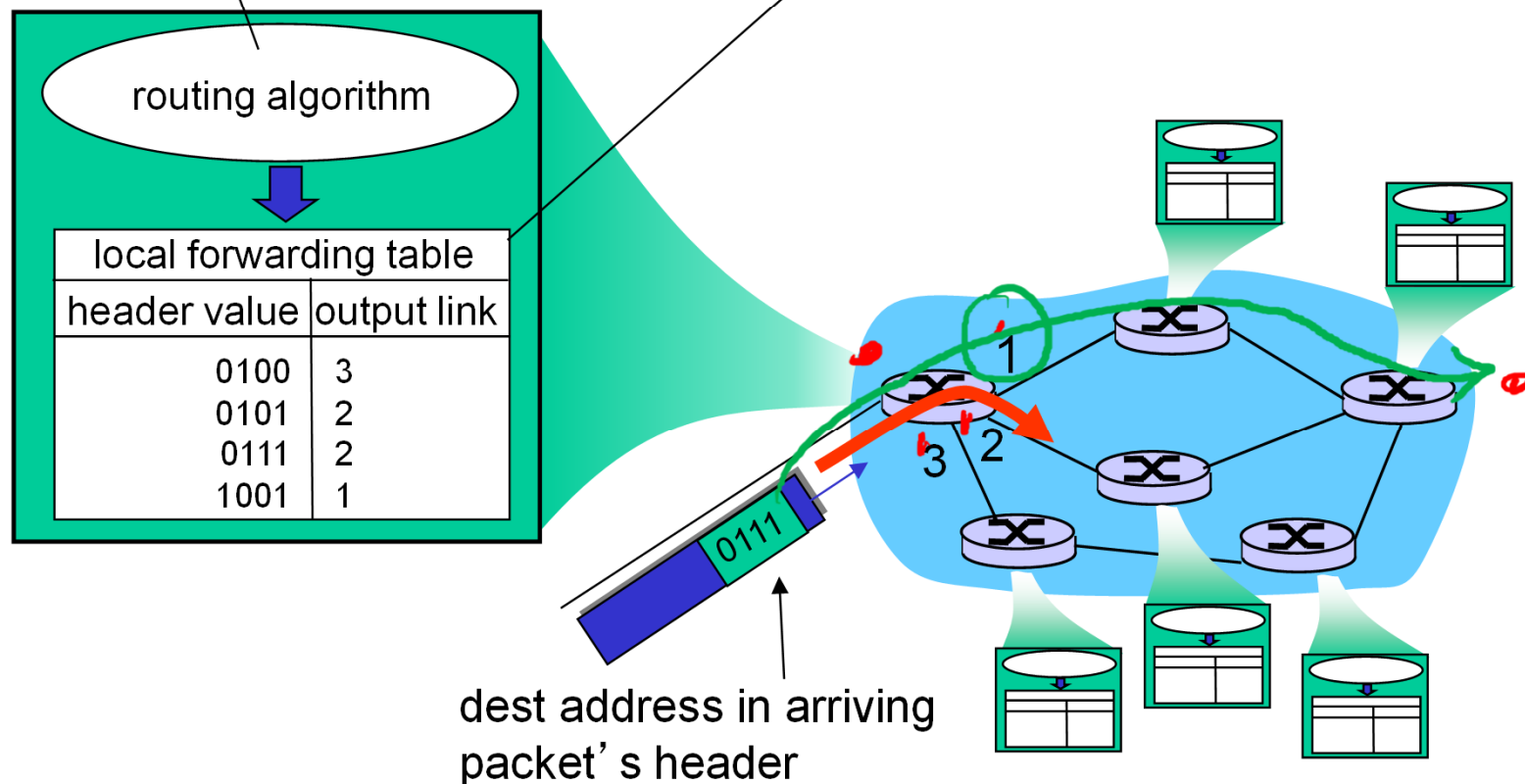
Two key network-core functions

end-to-end
routing: determines source-destination route taken by packets

- *global* routing algorithms

forwarding: move packets from router's input to appropriate router output

local

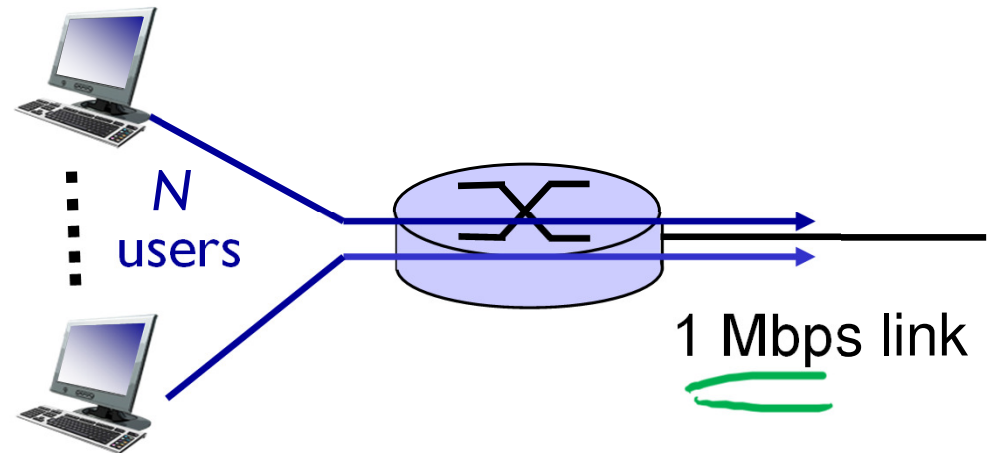


Packet switching versus circuit switching

packet switching allows more users to use network!

example:

- 1 Mb/s link
- each user:
 - 100 kb/s when “active”
 - active 10% of time



❖ *circuit-switching:*

- 10 users

$$= \frac{\text{total resource}}{\text{resource consumed by one user}}$$

$$= \frac{1 \text{ Mb/s}}{100 \text{ kb/s}} = \frac{1 \times 10^6}{100 \times 10^3}$$

❖ *packet switching:*

- with 35 users, probability > 10 active at same time is less than .0004 *

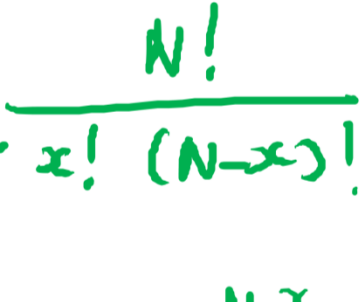
Binomial distribution

N : total # users ($N=35$)

p : active prob ($p=0.1$)

n : capacity ($n=10$)

$$f(x) = P(x \text{ simultaneous users}) = \binom{N}{x} p^x (1-p)^{N-x} \dots (1)$$



$$P(\text{overloaded}) = P(>n \text{ simultaneous users})$$

$$= f(n+1) + f(n+2) + \dots + f(N)$$

$$= \sum_{x=n+1}^N \binom{N}{x} p^x (1-p)^{N-x}$$

Packet switching versus circuit switching

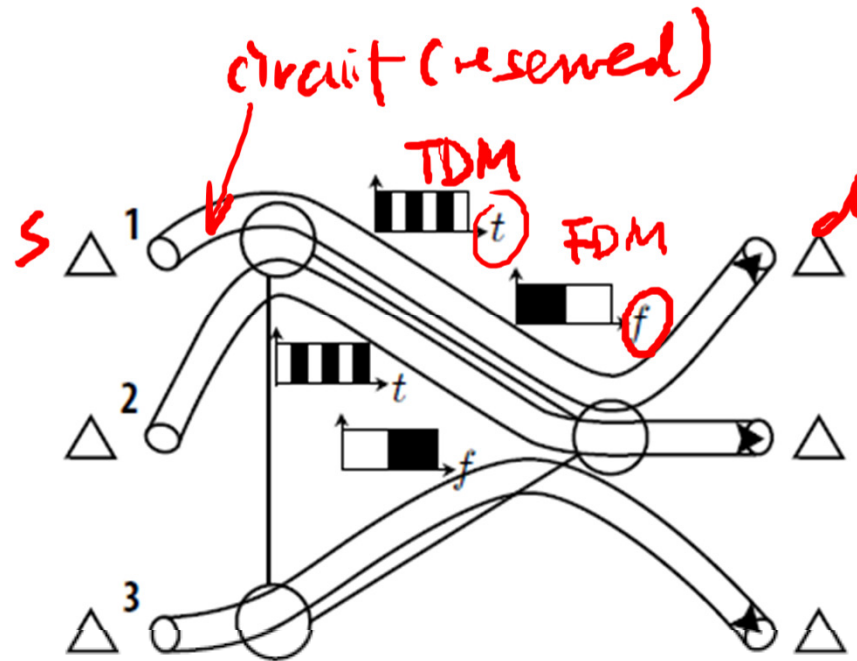
Is packet switching a “slam dunk winner?”

For packet switching:

- ❖ Adv: great for bursty data (eg $P=0.1$)
 - resource sharing
 - simpler, no call setup (eg $P=0.9$)
- ❖ Dis: excessive congestion possible → packet delay and loss
 - protocols needed for reliable data transfer, congestion control

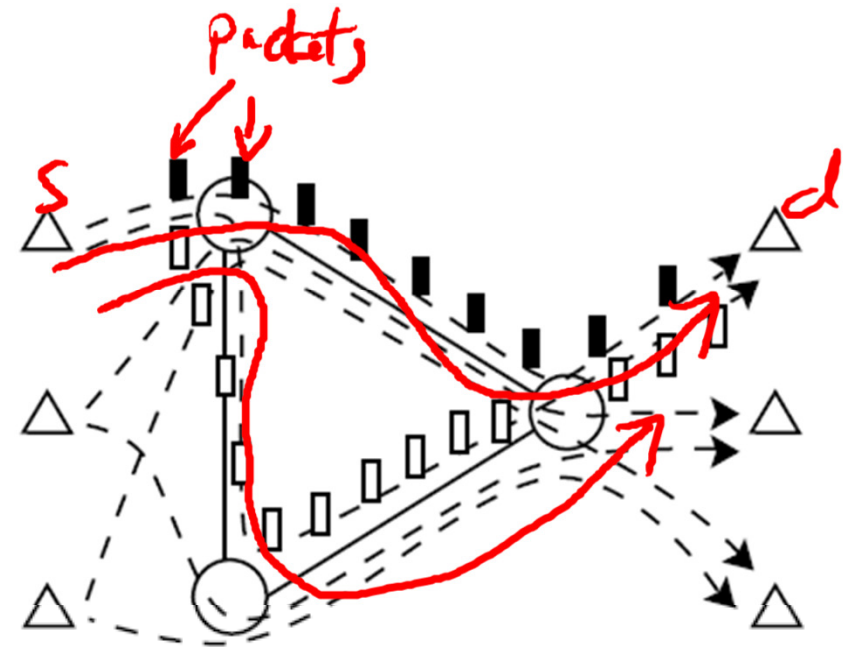
reserved resources (circuit switching) versus on-demand allocation (packet switching)

Circuit switching versus Packet switching



(a)

circuit switching



(b)

packet switching

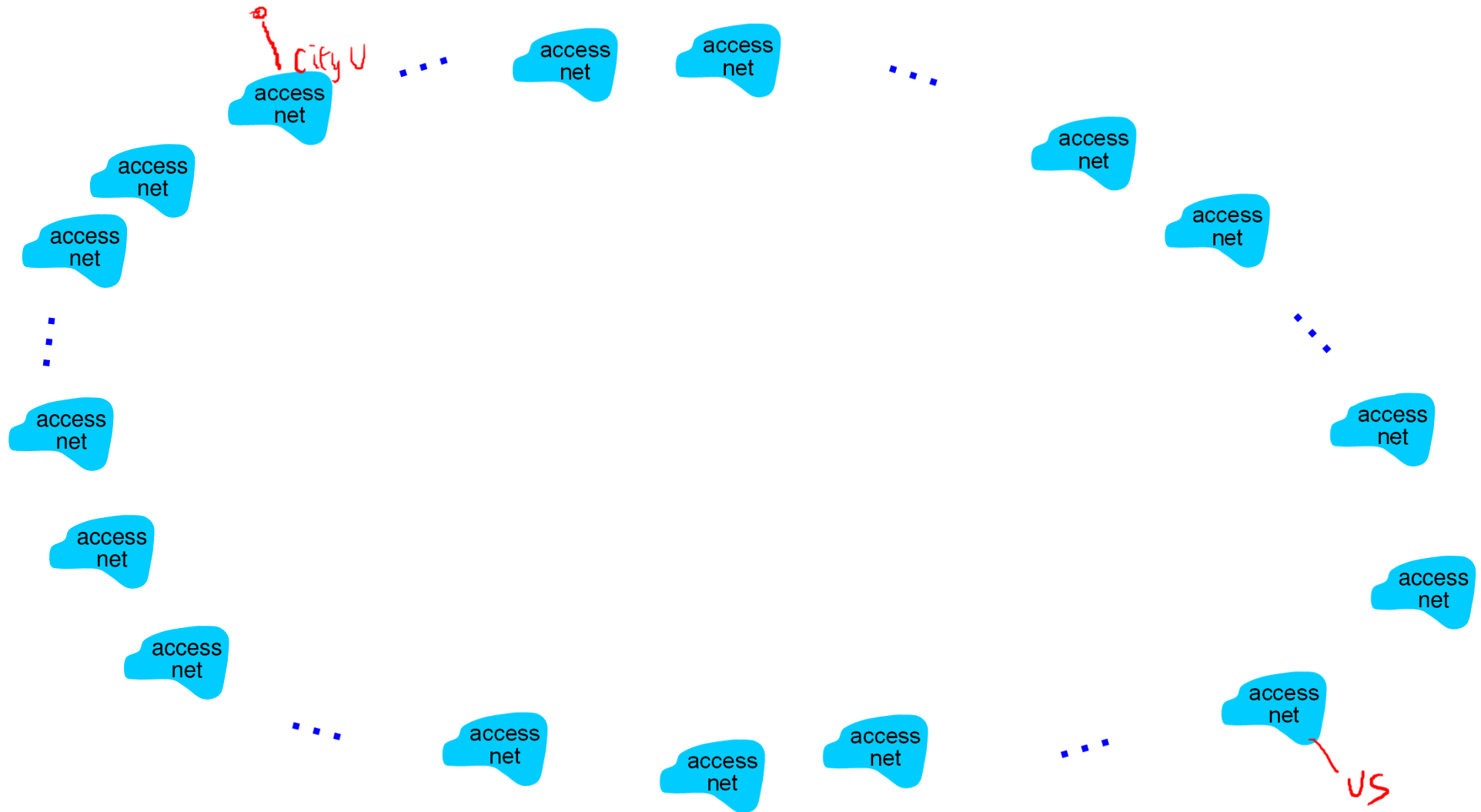
Internet structure

Internet structure: network of networks

- ❖ End systems connect to Internet via access ISPs (Internet Service Providers)
 - Residential, company and university ISPs
- ❖ Access ISPs in turn must be interconnected.
 - ❖ So that any two hosts can send packets to each other
- ❖ Resulting network of networks is very complex
 - ❖ Evolution was driven by economics and national policies
- ❖ Let's take a stepwise approach to describe current Internet structure

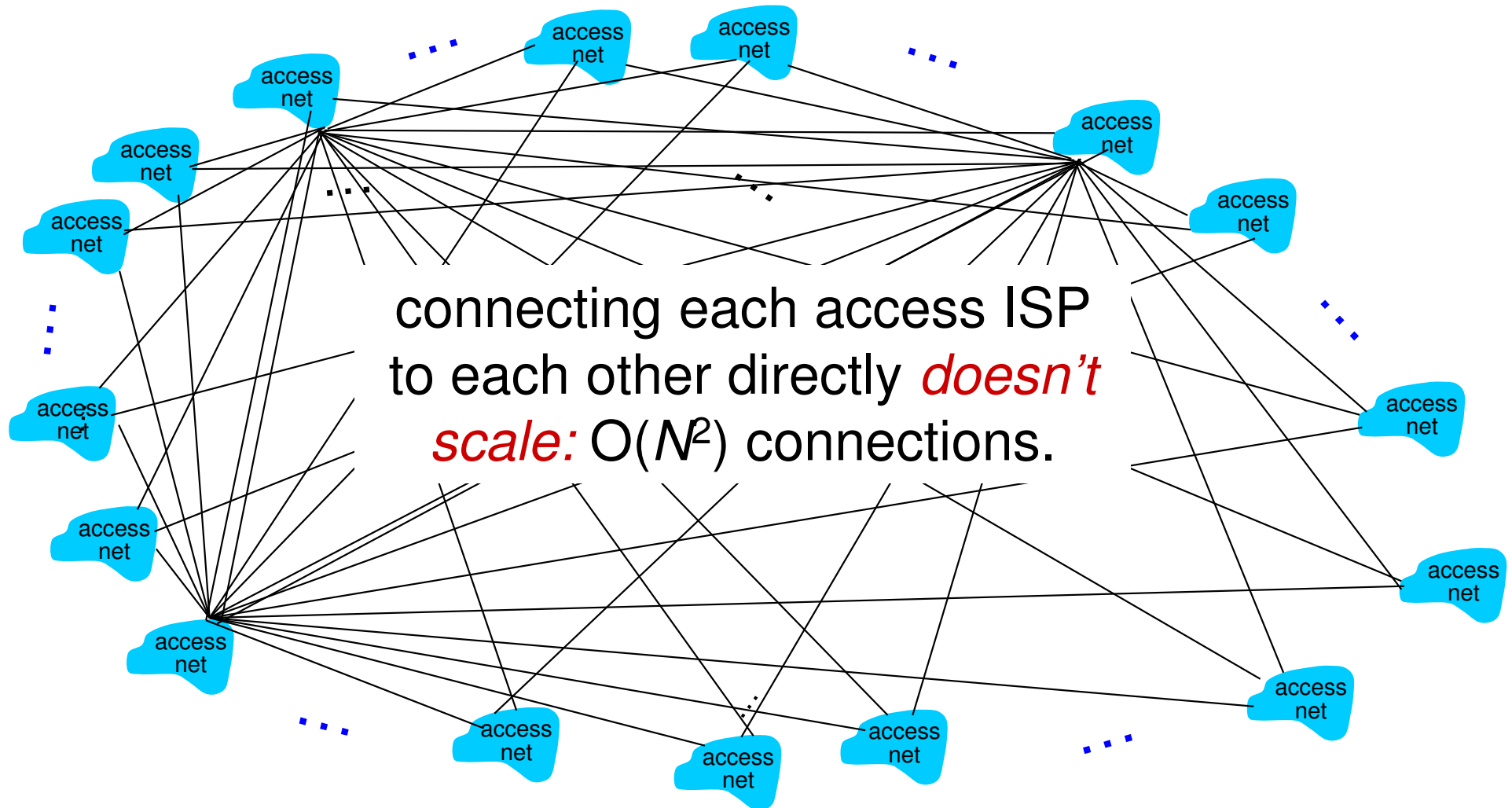
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



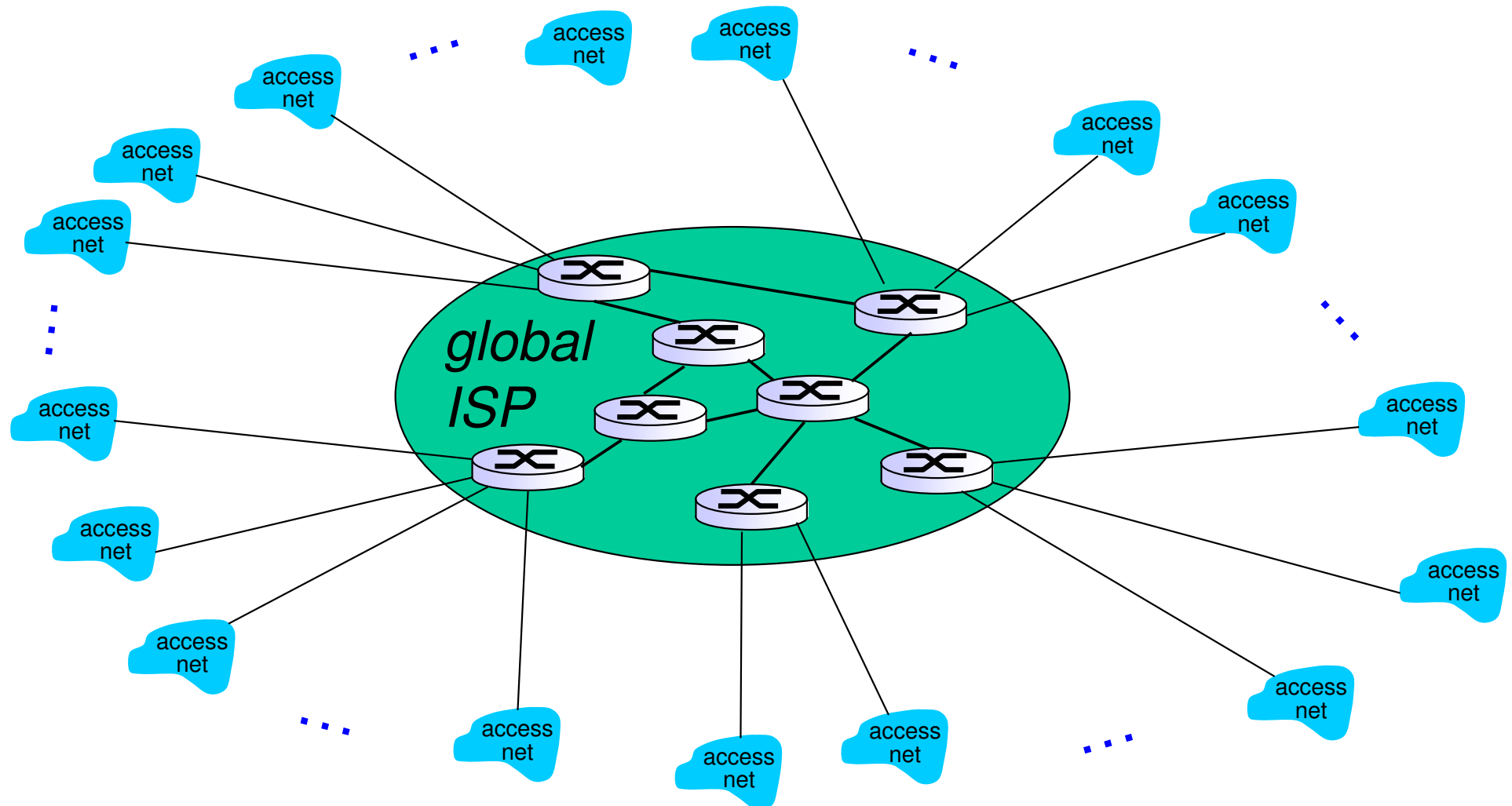
Internet structure: network of networks

Option: connect each access ISP to every other access ISP?



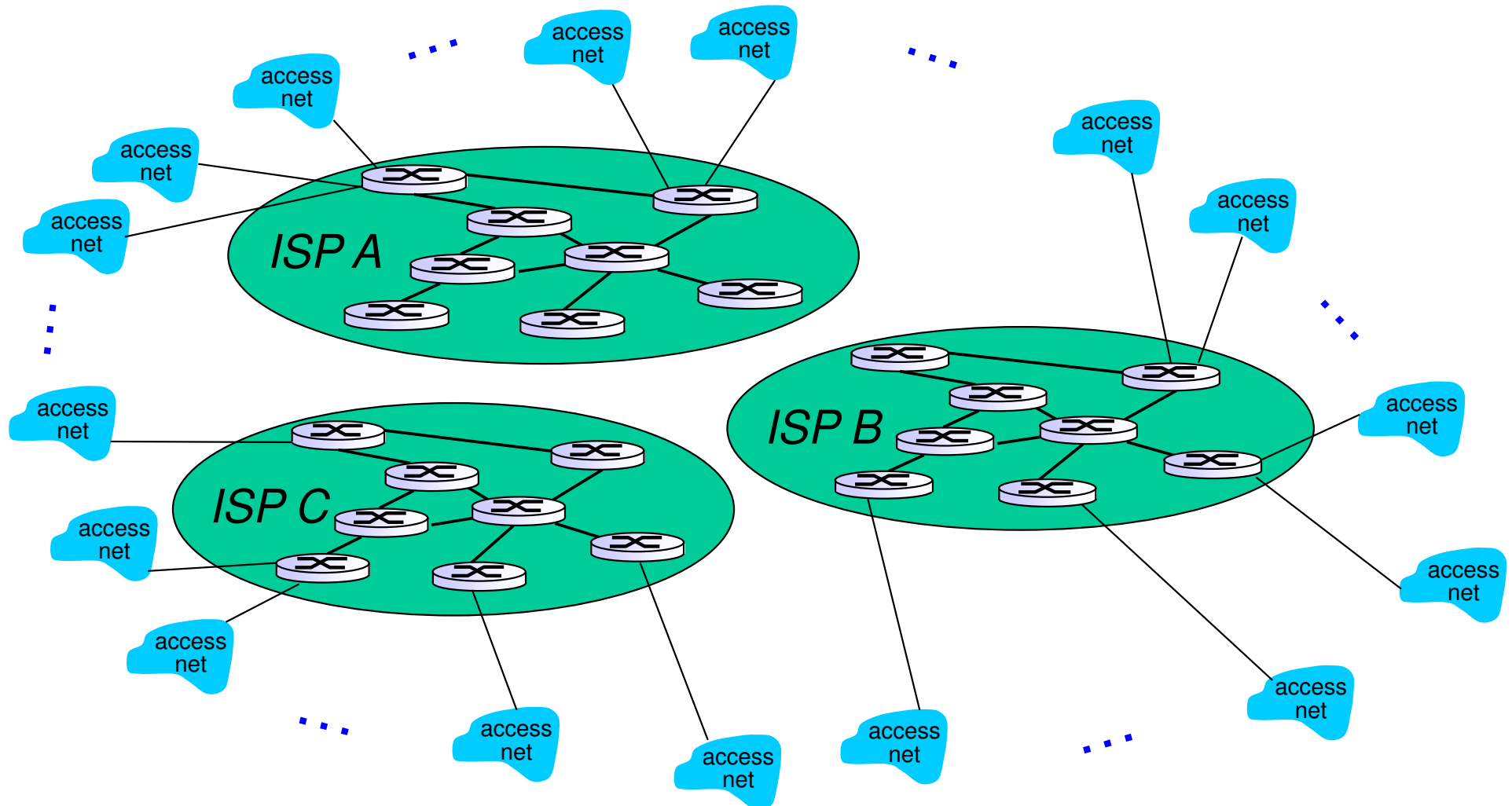
Internet structure: network of networks

*Option: connect each access ISP to a global transit ISP? **Customer** and **provider** ISPs have economic agreement.*



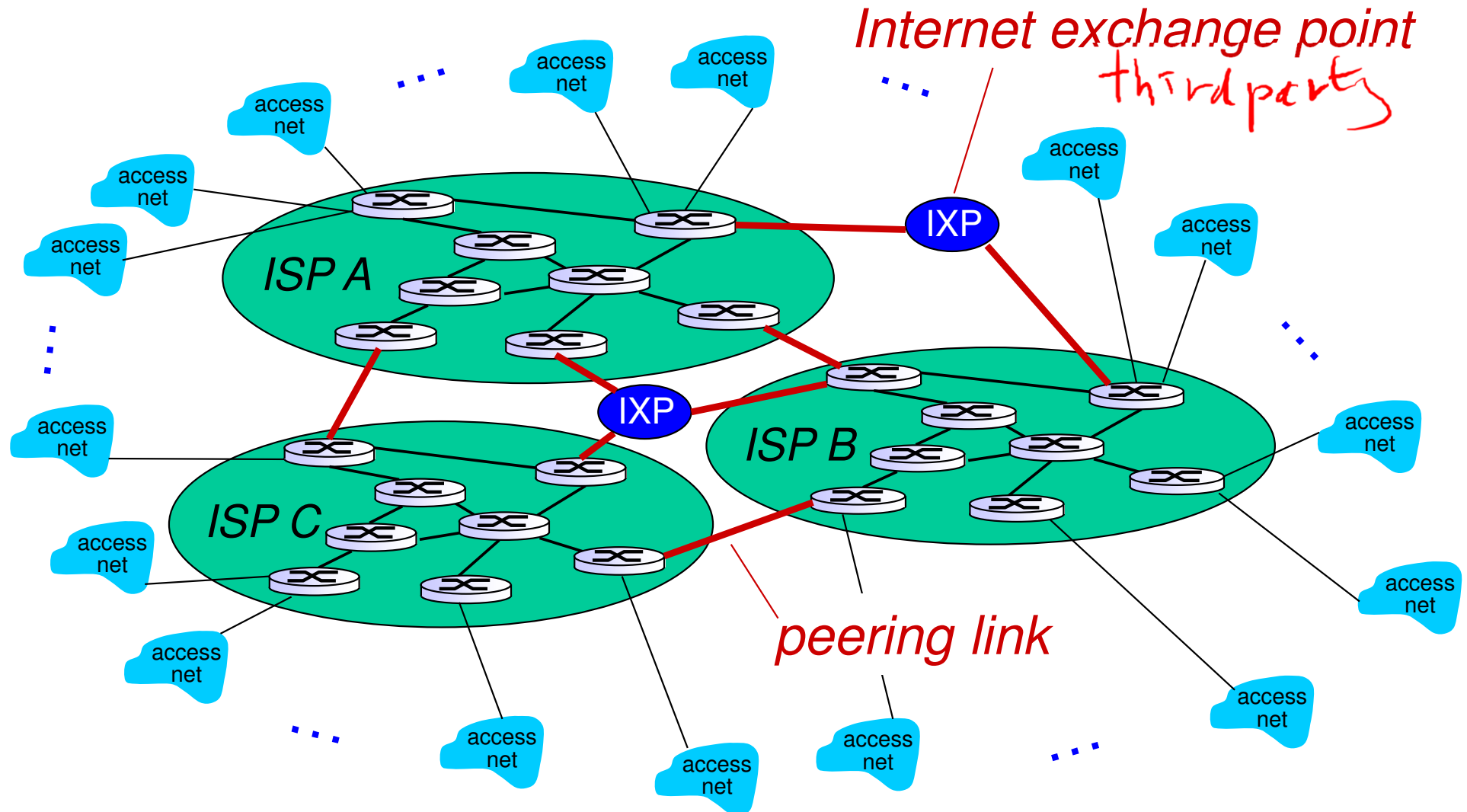
Internet structure: network of networks

But if **one global ISP** is viable business, there will be competitors



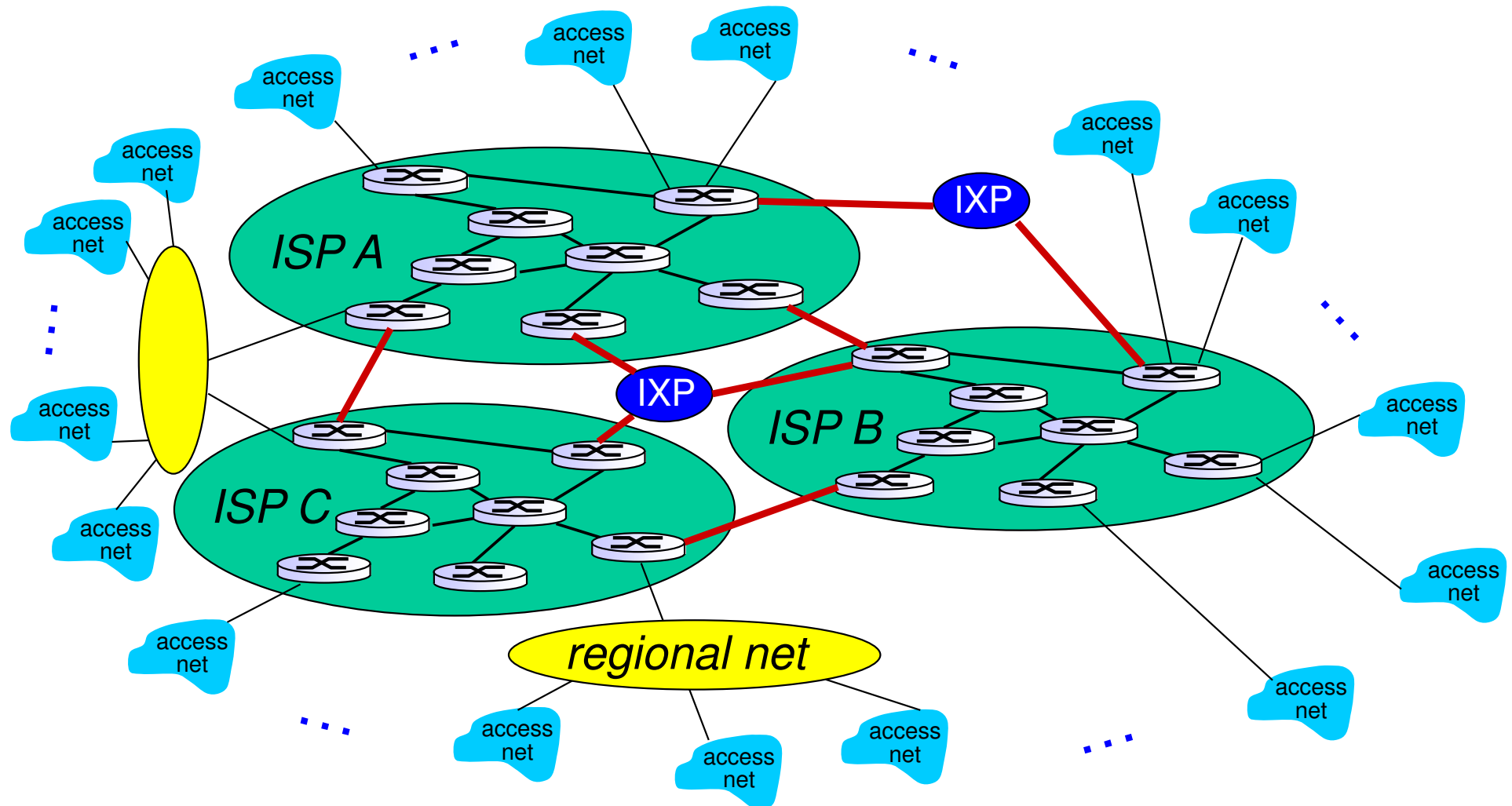
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors which must be interconnected



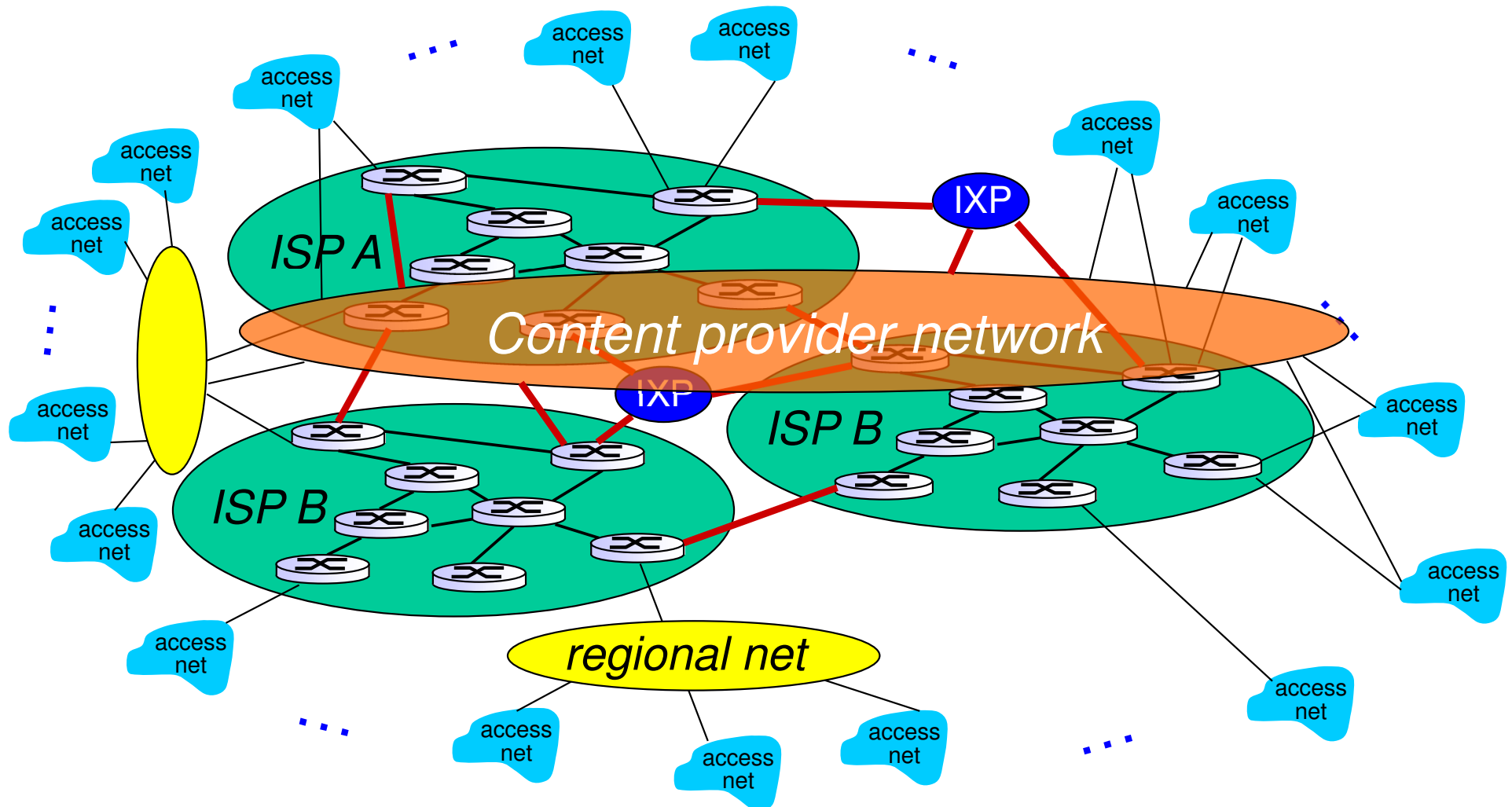
Internet structure: network of networks

... and regional networks may arise to connect access nets to ISPs

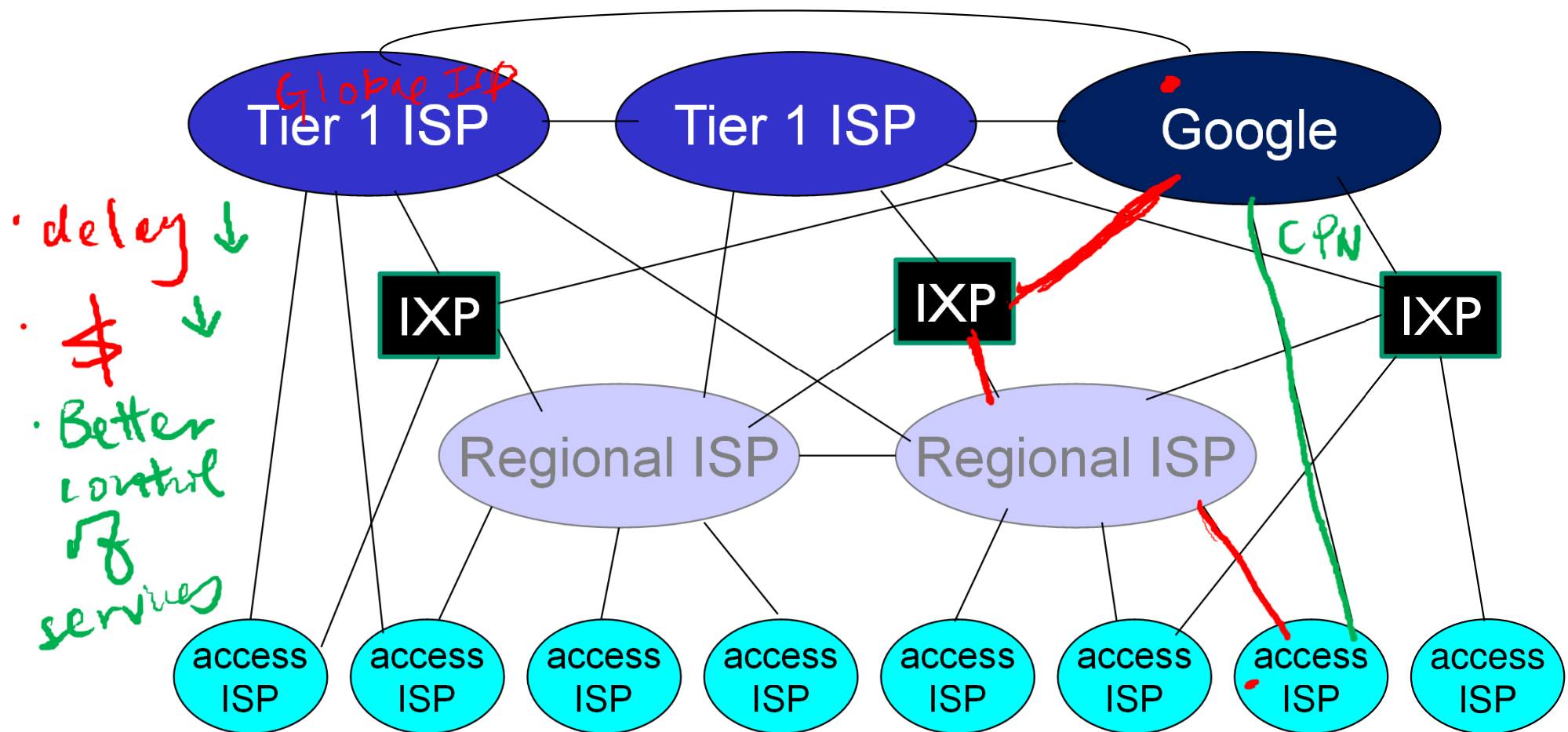


Internet structure: "network of networks"

... and **content provider networks** (e.g., Google, Microsoft, Akamai) may run their **own network**, to bring services, content close to end users



Internet structure: network of networks!



❖ Small # of well-connected large networks

- “tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider network (e.g., Google): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

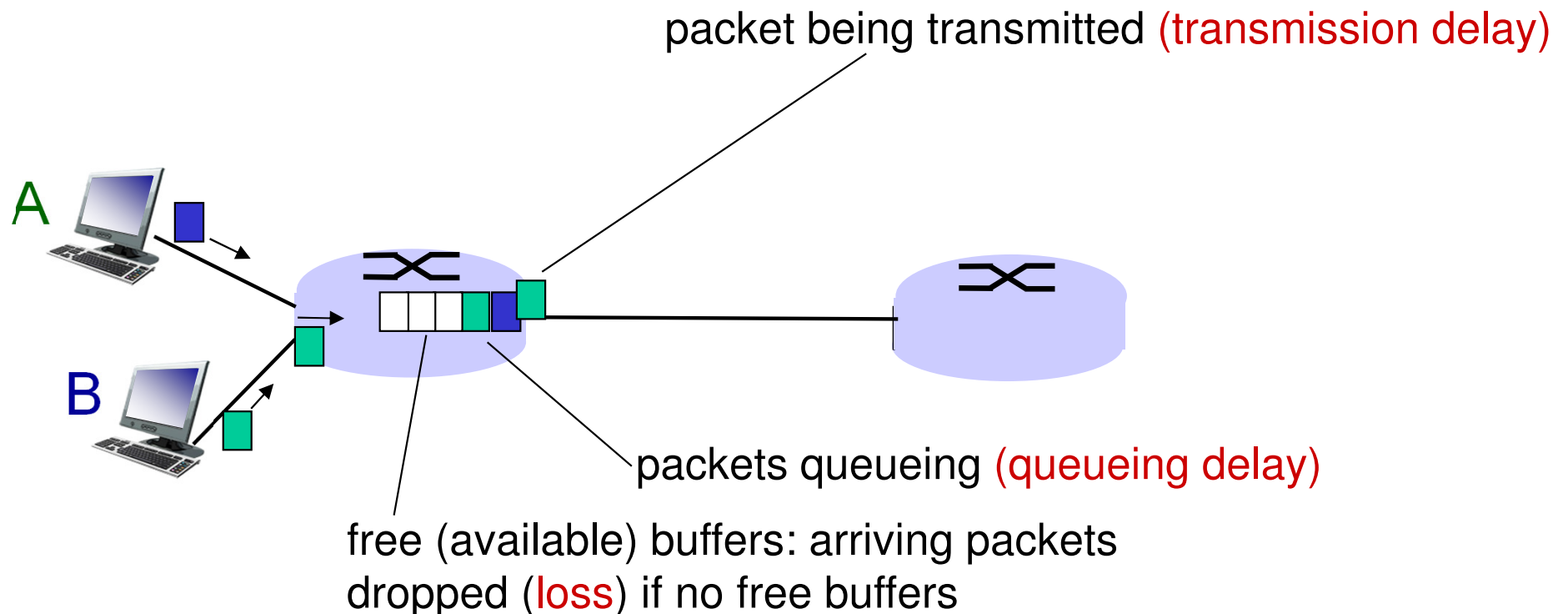
1.6 networks under attack: security

1.7 history

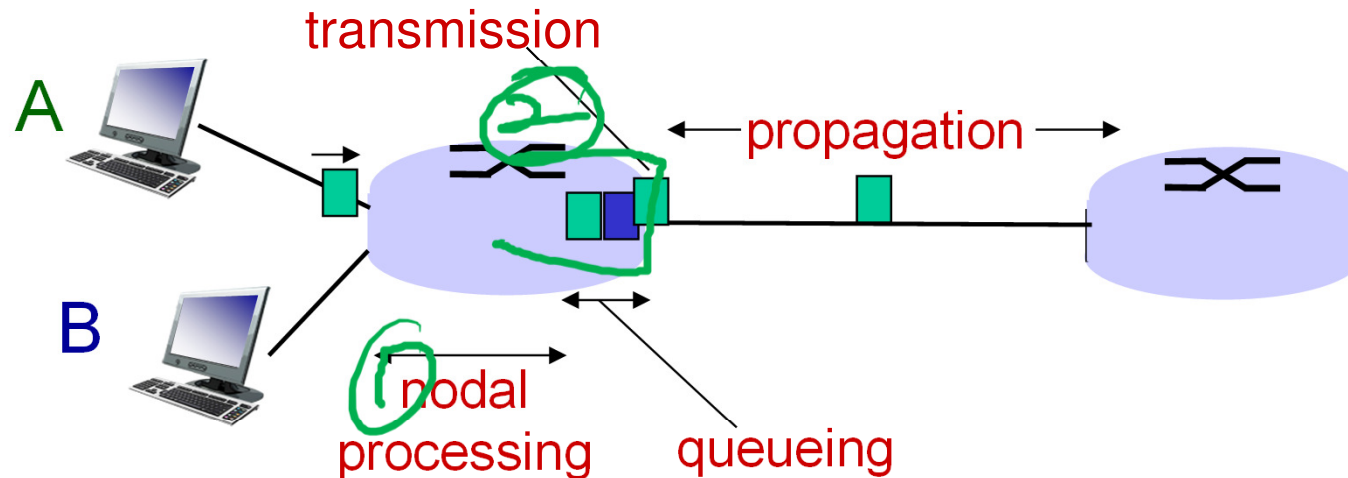
How do loss and delay occur?

packets *queue* in router buffers

- ❖ packet arrival rate to link (temporarily) exceeds output link capacity
- ❖ packets queue, wait for turn



Four sources of packet delay



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

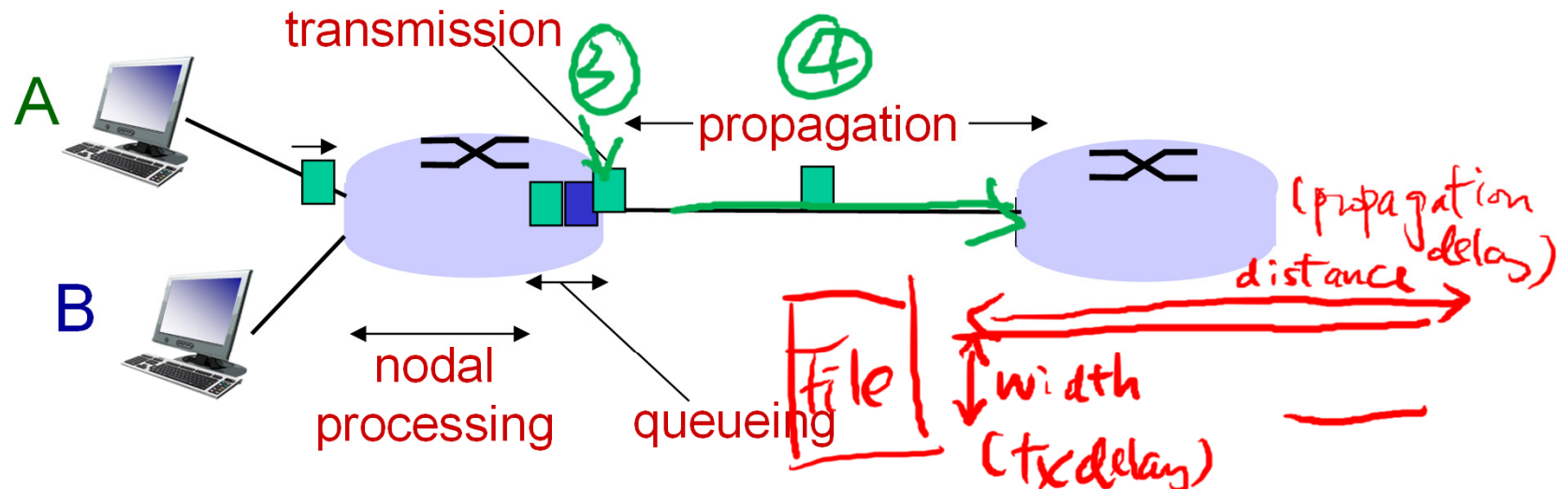
d_{proc} : ① nodal processing

- check bit errors
- determine output link
- typically < msec

d_{queue} : ② *Can be very significant* queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Four sources of packet delay



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

⑤ d_{trans} : transmission delay:

- L : packet length (bits)
- R : link bandwidth (bps)
- $d_{\text{trans}} = L/R$

④ d_{prop} : propagation delay:

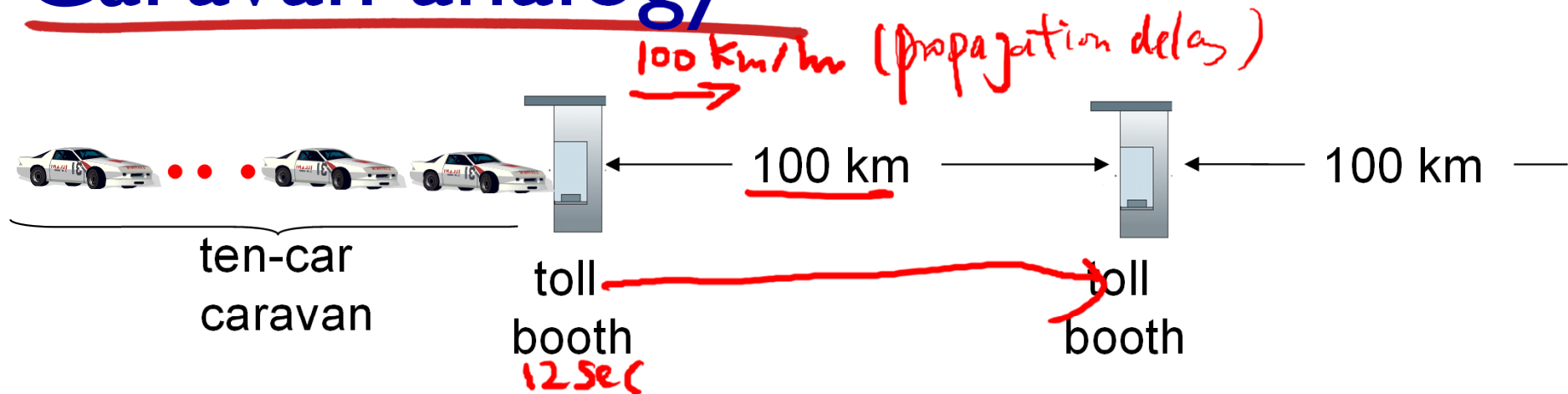
- d : length of physical link
- s : propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- $d_{\text{prop}} = d/s$

d_{trans} and d_{prop}
very different

* Check out the Java applet for an interactive animation on trans vs. prop delay:

https://wps.pearsoned.com/ecs_kurose_compnetw_6/216/55463/14198702.cw/index.html

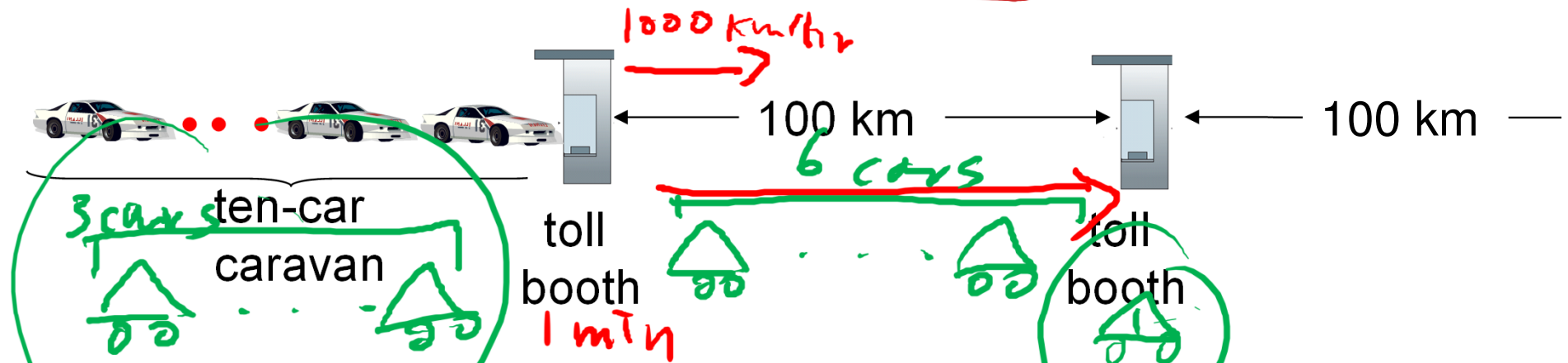
Caravan analogy



- ❖ cars “propagate” at 100 km/hr
- ❖ toll booth takes 12 sec to service car (bit transmission time)
- ❖ car ~ bit; caravan ~ packet
- ❖ Q: How long until caravan is lined up before 2nd toll booth?

- time to “push” entire caravan through toll booth onto highway = $12 * 10 = 120$ sec = 2 min
- time for last car to propagate from 1st to 2nd toll booth: $100\text{km} / (100\text{km/hr}) = 1 \text{ hr}$
- A: 62 minutes

Caravan analogy (more)



- ❖ suppose cars now “propagate” at 1000 km/hr
- ❖ and suppose toll booth now takes 1 min to service a car
- ❖ Q: Will cars arrive to 2nd booth before all cars serviced at first booth?

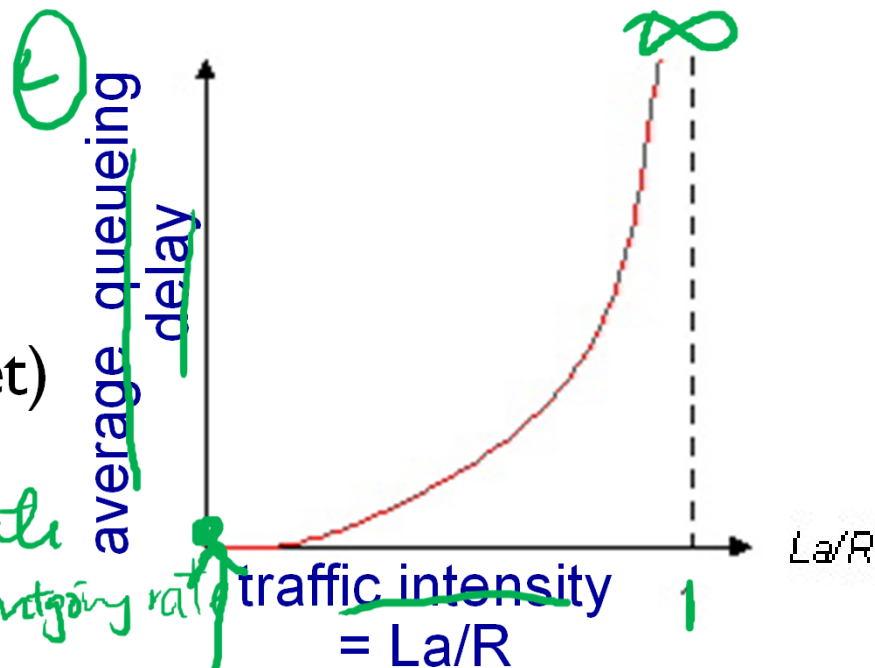
Yes

$$\text{Time for 1st car to reach 2nd booth} \\ = 1 \text{ min} + \frac{100 \text{ km}}{1000 \text{ km/hr}} = 7 \text{ min}$$

Queueing delay (revisited)

- ❖ R : link bandwidth (bps = bits/sec)
- ❖ L : packet length (bits/packet)
- ❖ a : average packet arrival rate (packets/sec)

traffic intensity = $\frac{La}{R}$



- ❖ $\frac{La}{R} \sim 0$: avg. queueing delay small
- ❖ $\frac{La}{R} \rightarrow 1$: avg. queueing delay large
- ❖ $\frac{La}{R} > 1$: more “work” arriving than can be serviced, average delay infinite!



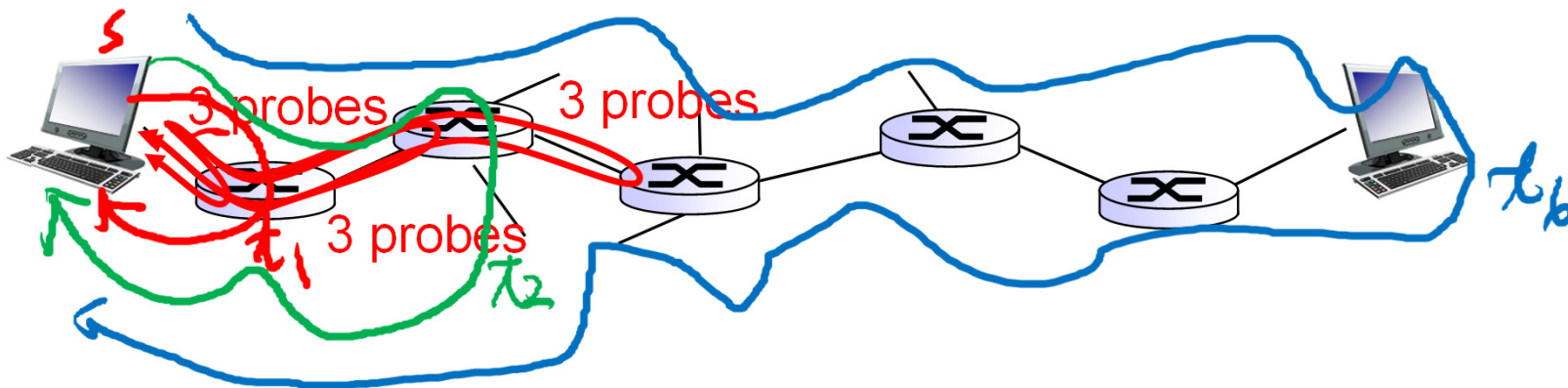
$\frac{La}{R} \sim 0$



$\frac{La}{R} \rightarrow 1$

“Real” Internet delays and routes

- ❖ what do “real” Internet delay & loss look like?
- ❖ traceroute program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination
 - router i will return packets to sender
 - sender times interval between transmission and reply.



“Real” Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

→ 1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
→ 2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms

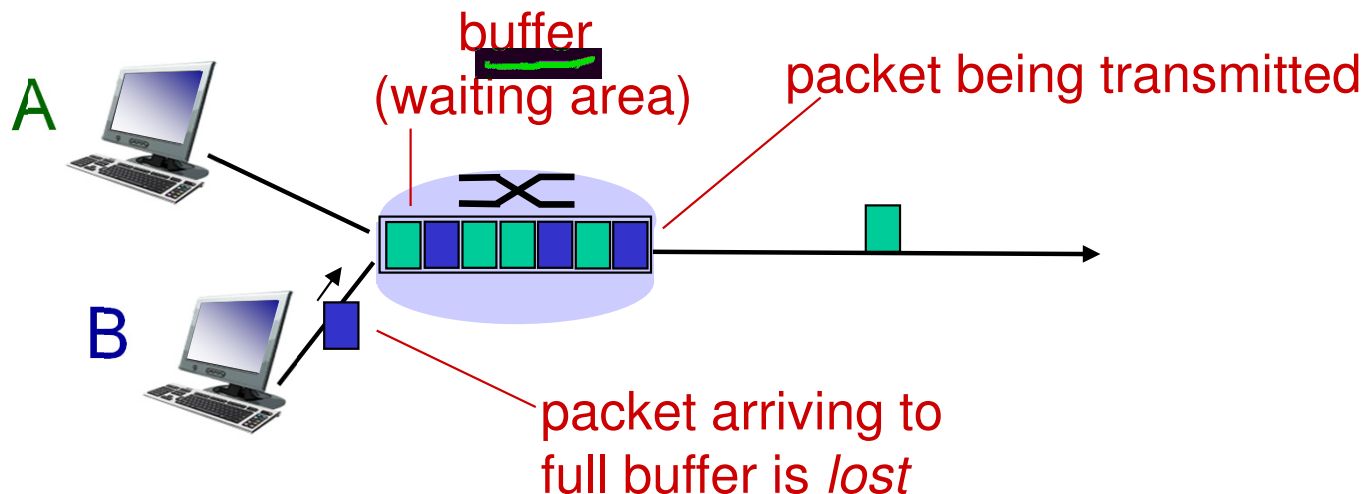
trans-oceanic
link

* means no response (probe lost, router not replying)

19 hops dest

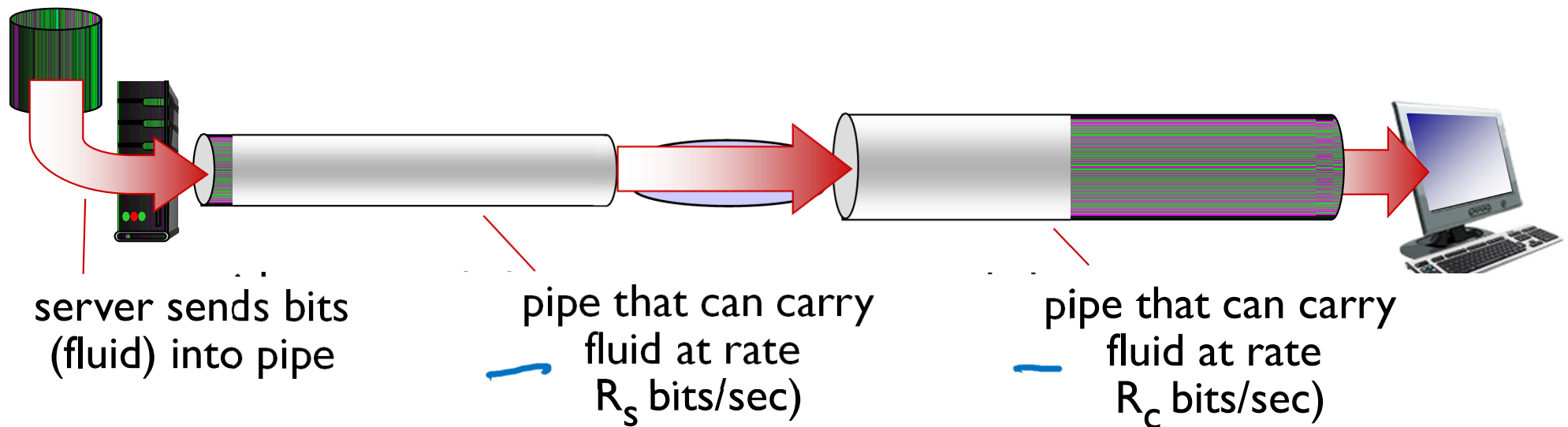
Packet loss

- ❖ queue (aka buffer) preceding link in buffer has **finite capacity**
- ❖ packet arriving to full queue **dropped** (aka **lost**)
- ❖ lost packet may be retransmitted by previous node, by source end system, or not at all



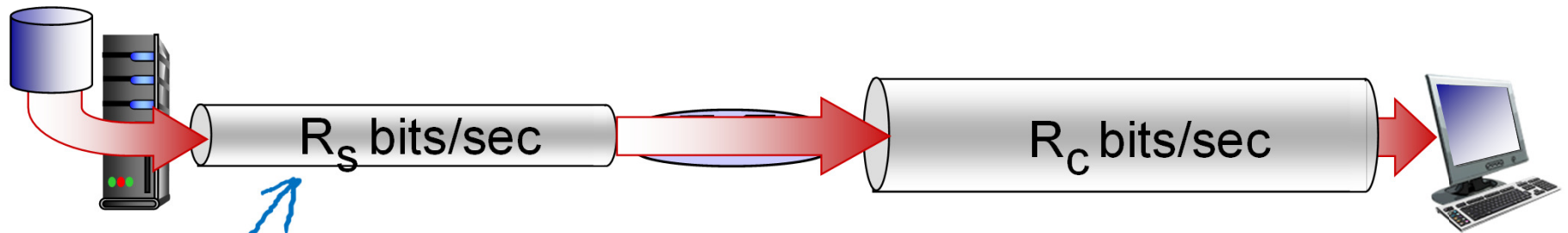
Throughput

- ❖ *throughput*: rate (bits/time unit) at which bits transferred between sender/receiver
 - *instantaneous*: rate at given point in time
 - *average*: rate over longer period of time

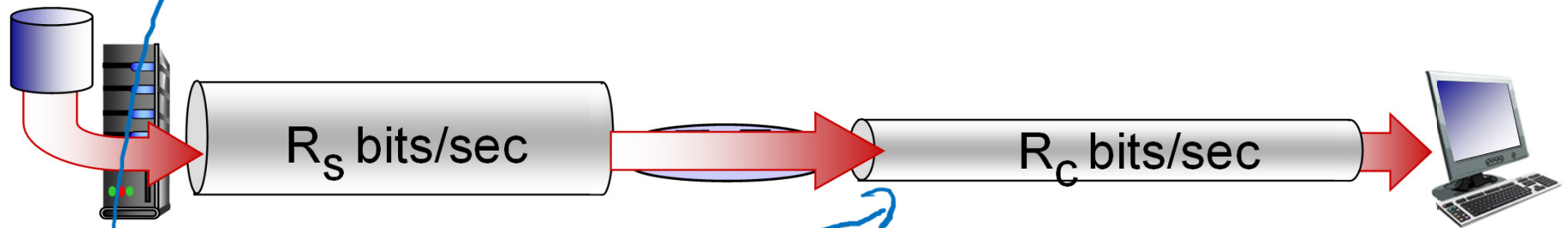


Throughput (more) $= \min \{ R_s, R_c \}$

❖ $R_s < R_c$ What is average end-end throughput? R_s



❖ $R_s > R_c$ What is average end-end throughput? R_c

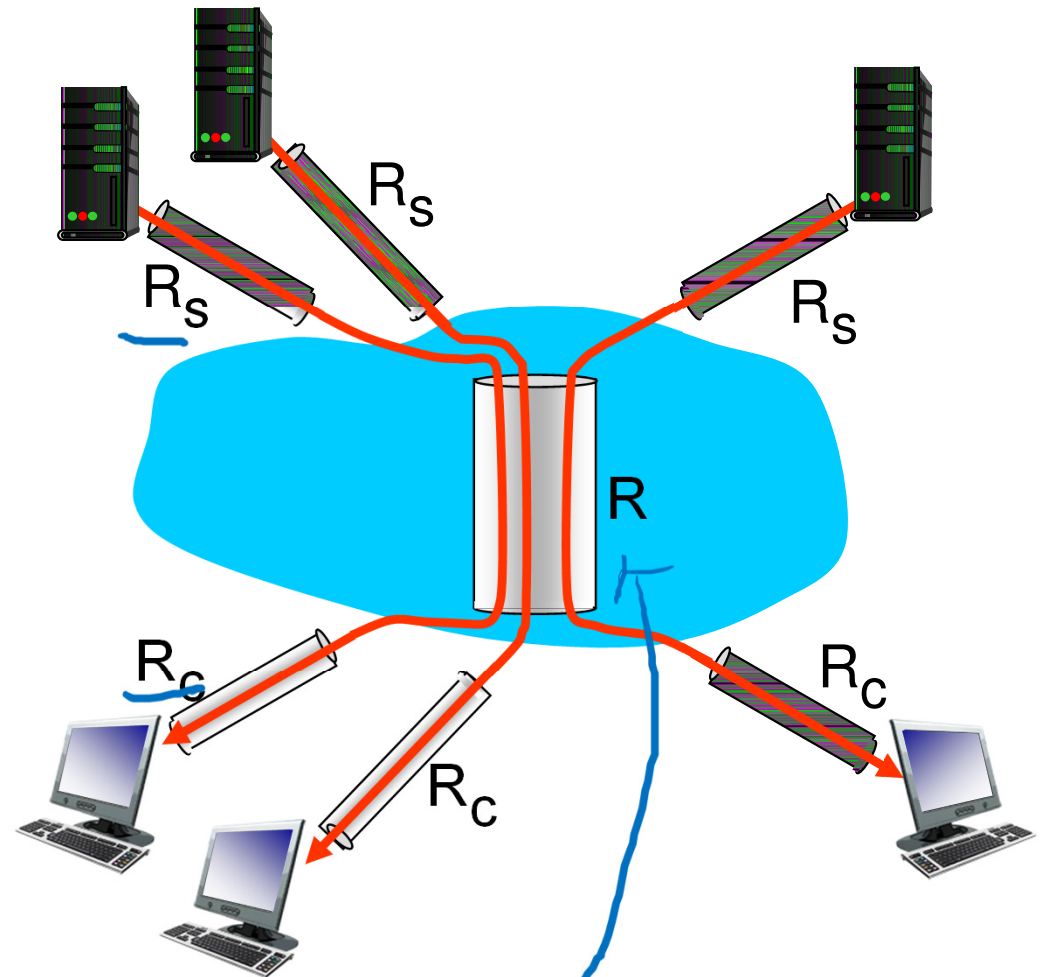


bottleneck link

link on end-end path that constrains end-end throughput

Throughput: Internet scenario

- ❖ per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- ❖ in practice: R_c or R_s is often bottleneck



10 connections (fairly) share
backbone bottleneck link R bits/sec

Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

Protocol “layers”

*Networks are complex,
with many “pieces”:*

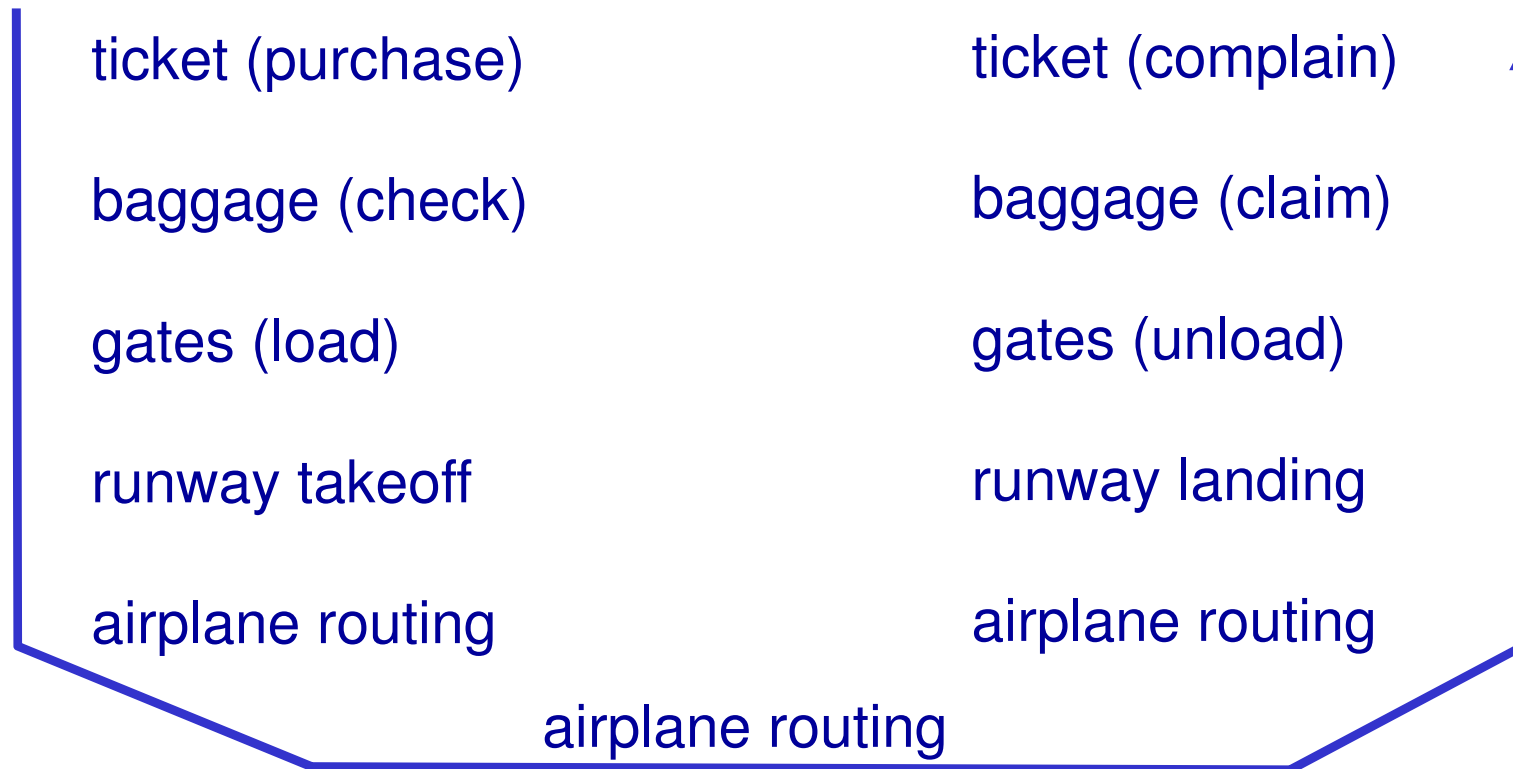
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question:

is there any hope of
organizing structure of
network?

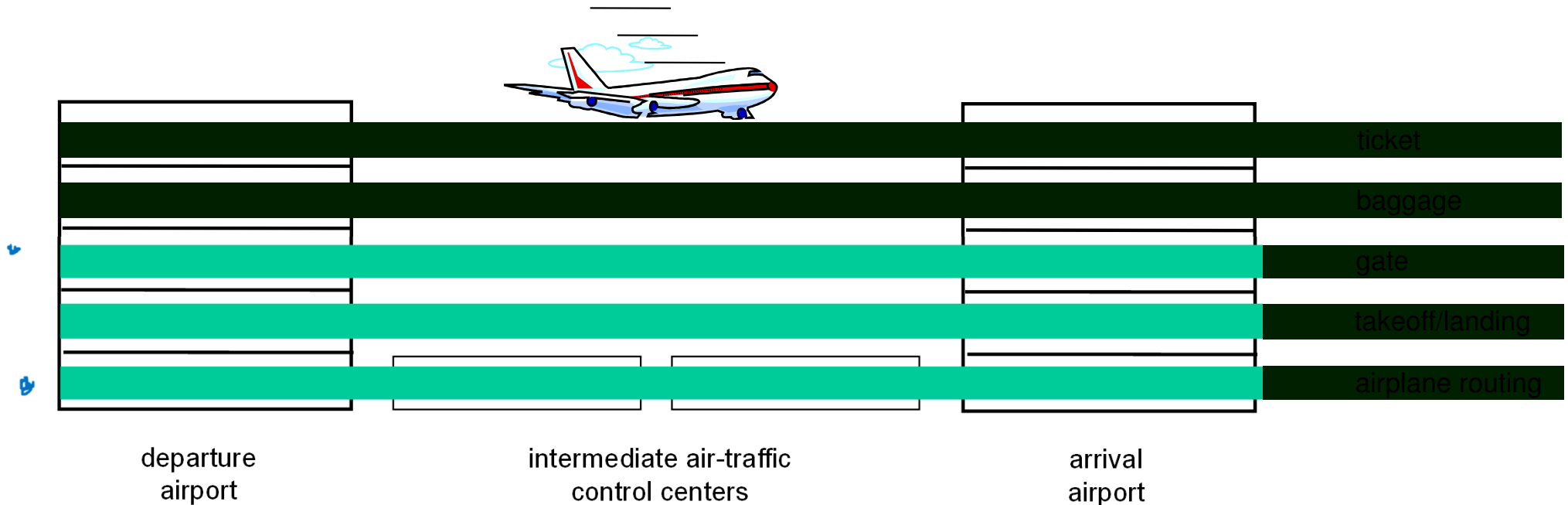
.... or at least our
discussion of networks?

Organization of air travel



❖ a series of steps

Layering of airline functionality



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below
 - E.g., At the ticketing layer and below: airline-counter-to-airline-counter transfer of a person

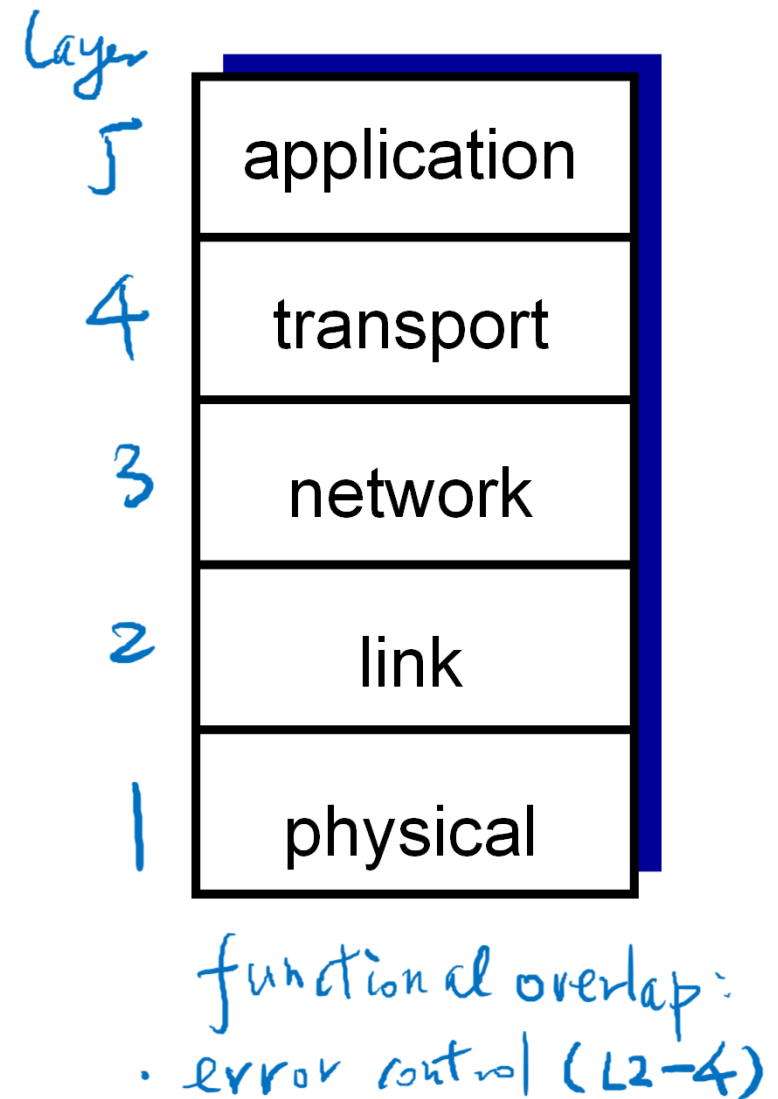
Why layering?

Dealing with **complex systems**:

- ❖ **modularization** eases maintenance and updating of system
 - change of implementation of layer's service transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system
- ❖ explicit structure allows **identification of relationship** among complex system's pieces
 - layered *reference model* for discussion

Internet protocol stack

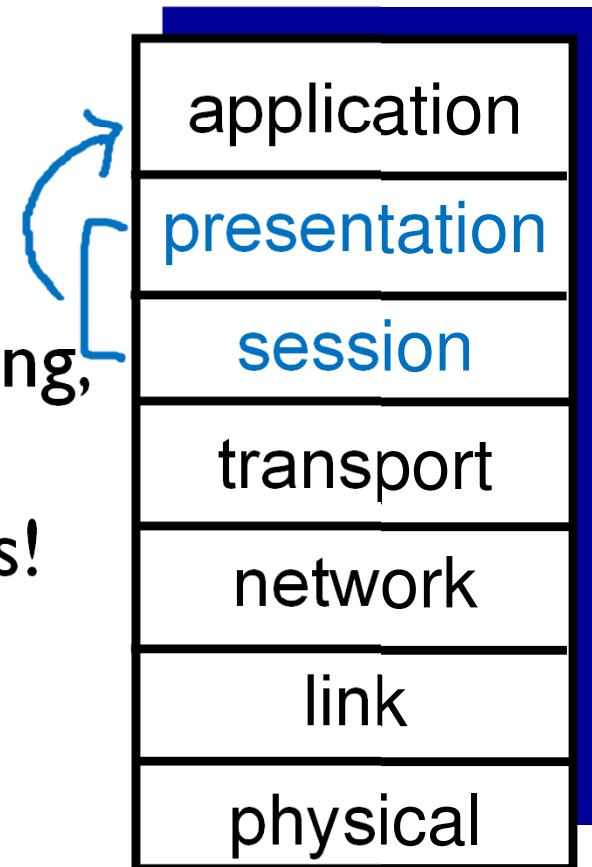
- ❖ *application*: supporting network applications
 - FTP, SMTP, HTTP
- ❖ *transport*: process-process data transfer
 - TCP, UDP
- ❖ *network*: routing of datagrams from source to destination
 - IP, routing protocols
- ❖ *link*: data transfer between neighboring network elements
 - Ethernet, 802.11 (Wi-Fi), PPP
- ❖ *physical*: bits “on the wire”



Open systems Interconnection ISO/OSI reference model

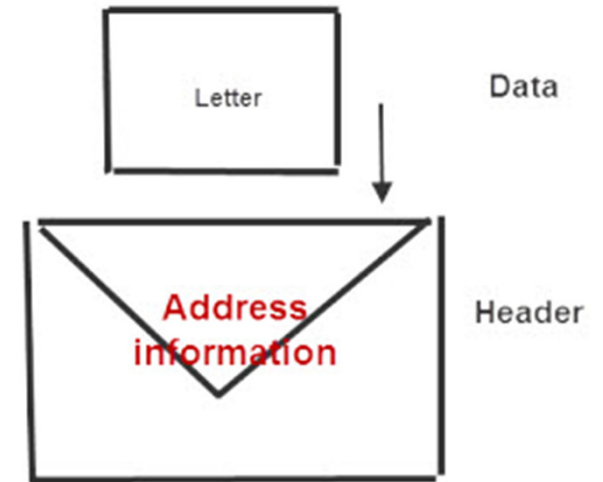
1970s

- ❖ **presentation**: allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- ❖ **session**: synchronization, checkpointing, recovery of data exchange
- ❖ Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in application
- ❖ Summary:
 - **OSI model**: An **idealized** model
 - ✓ ▪ **Internet protocol stack**: A **practical** implementation

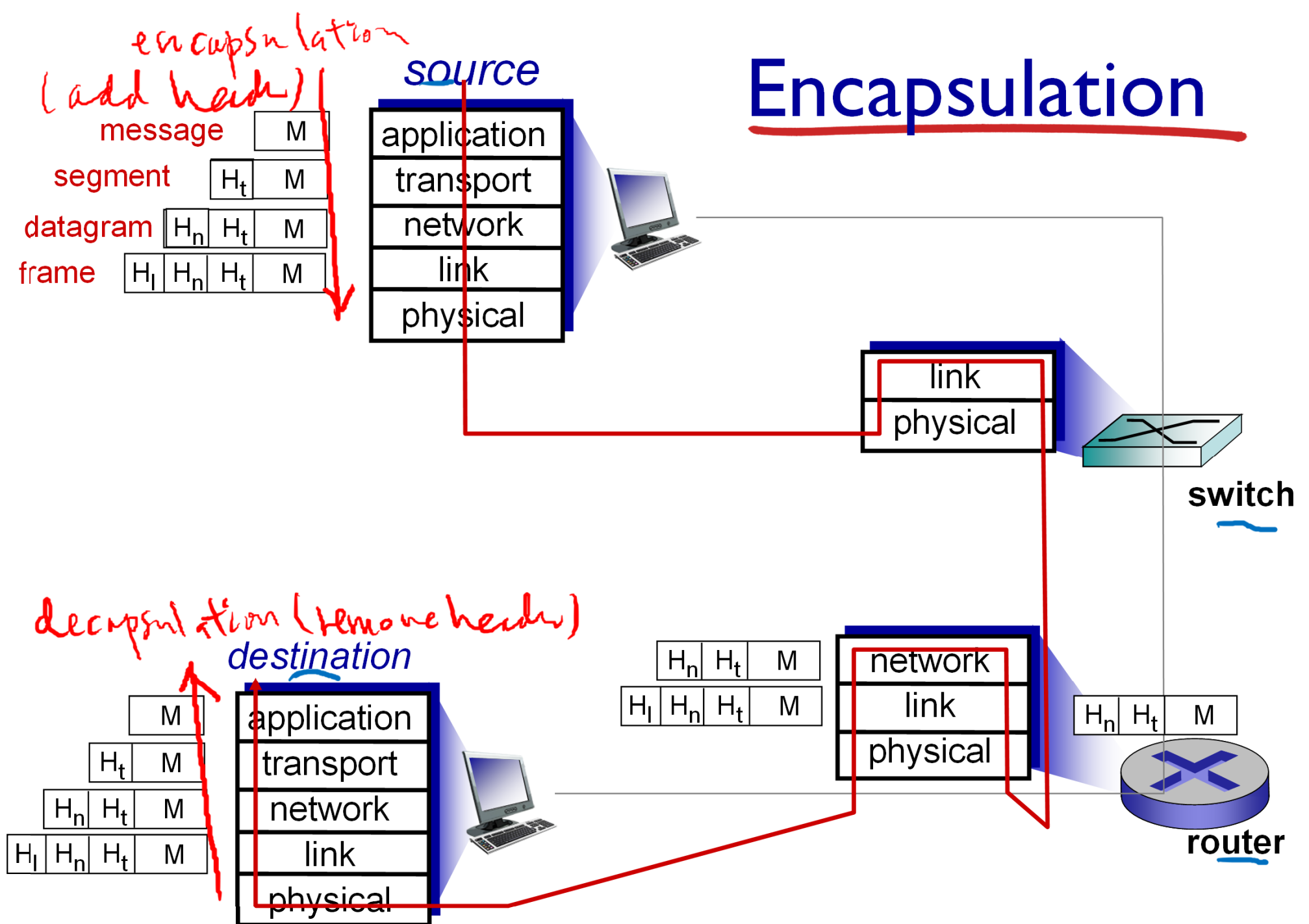


Encapsulation

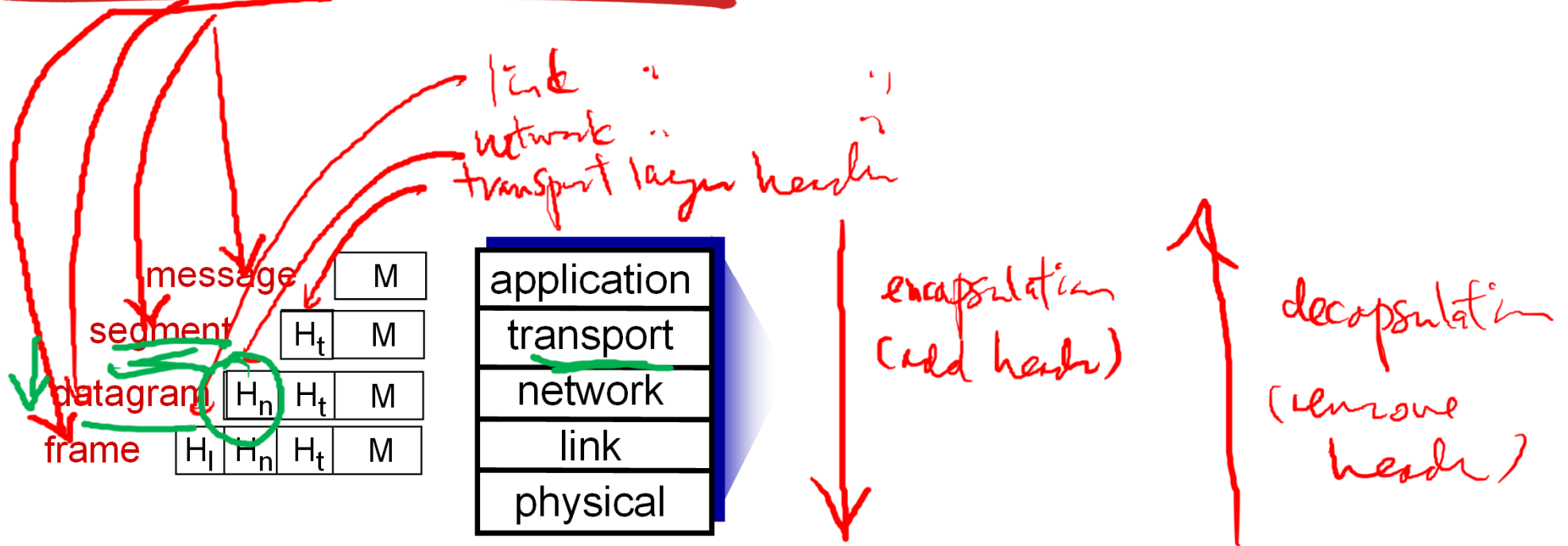
- ❖ **Encapsulation** takes information from a higher layer and adds a **header** to it, treating the higher layer information as data.



Encapsulation



“Packet” at different layers



Transpt layer: segment + H_n = datagram

Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

Network security

❖ field of network security:

- how bad guys can attack computer networks
- how we can defend networks against attacks
- how to design architectures that are immune to attacks

❖ Internet not originally designed with (much) security in mind!!

- *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
- Internet protocol designers playing “catch-up”
- security considerations in all layers!

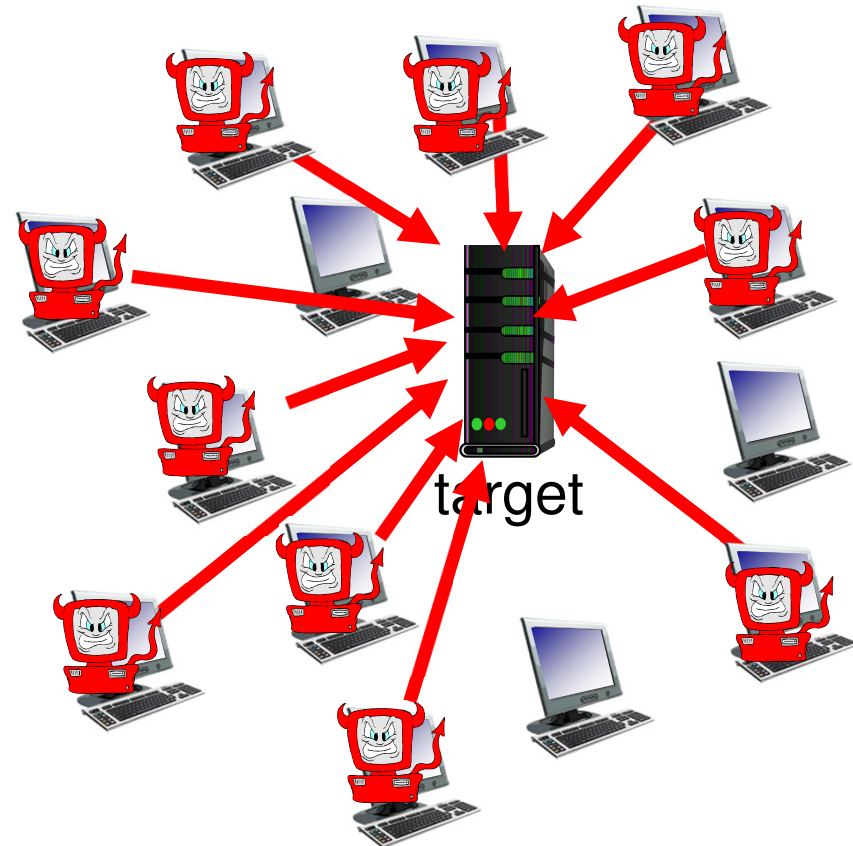
Bad guys: put malware into hosts via Internet

- ❖ malware can get in host from:
 - virus: self-replicating infection by receiving/executing object (e.g., e-mail attachment)
 - worm: self-replicating infection by passively receiving object that gets itself executed
- ❖ spyware can record keystrokes, web sites visited, upload info to collection site
- ❖ infected host can be enrolled in botnet, used for spam → DDoS attacks

Bad guys: attack server, network infrastructure

Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

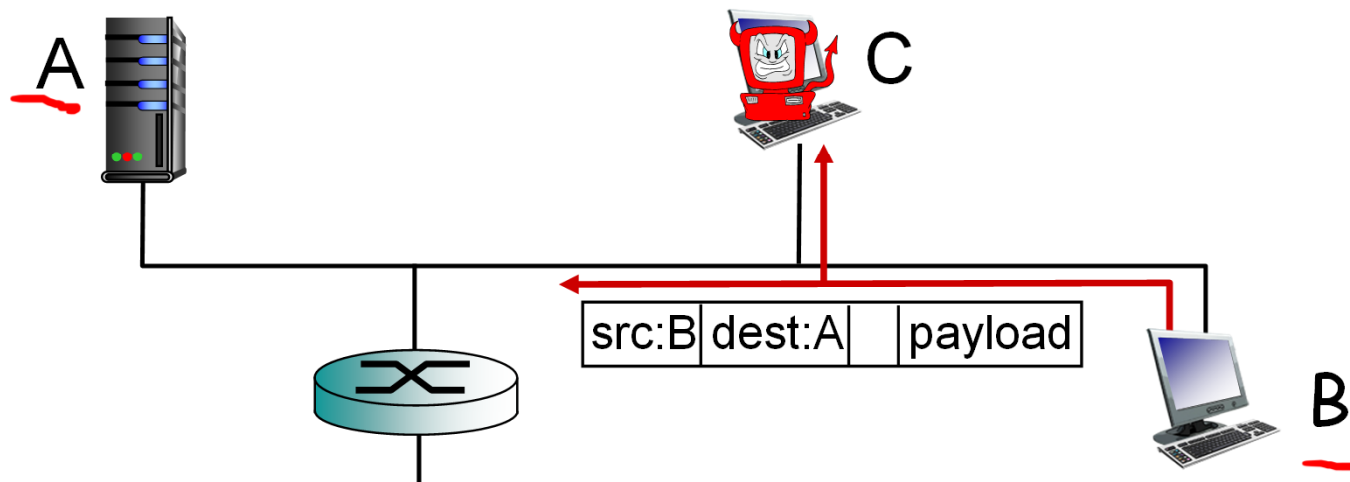
1. select target
2. break into hosts around the network
3. send packets to target from compromised hosts



Bad guys can sniff packets

Packet sniffing:

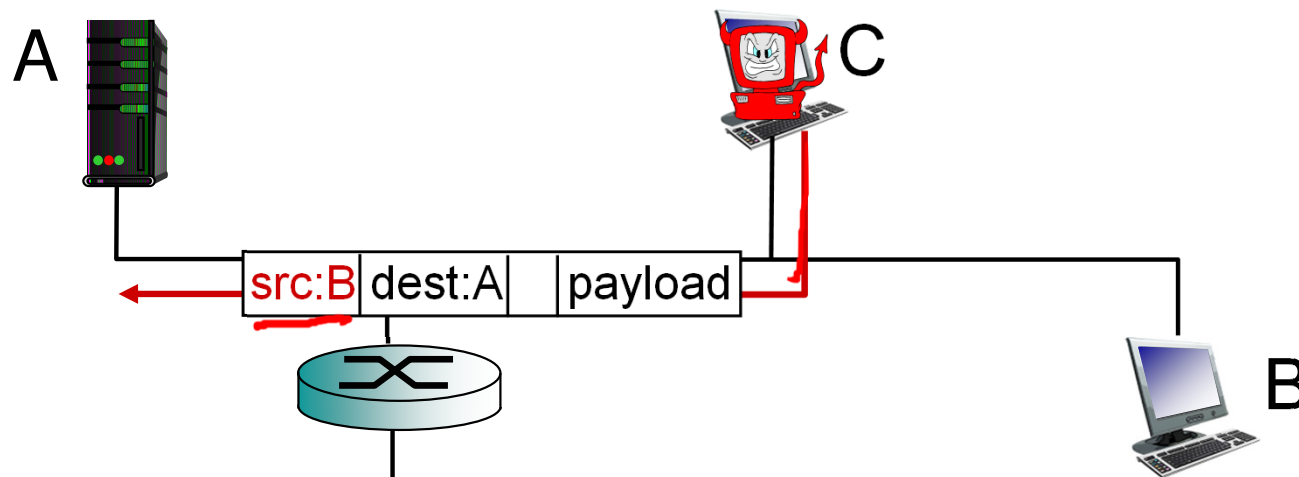
- broadcast media (shared ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



- ❖ wireshark software used for end-of-chapter labs is a (free) packet-sniffer

Bad guys can use fake addresses

IP spoofing: send packet with false source IP address



... lots more on security (throughout, Chapter 8)

Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

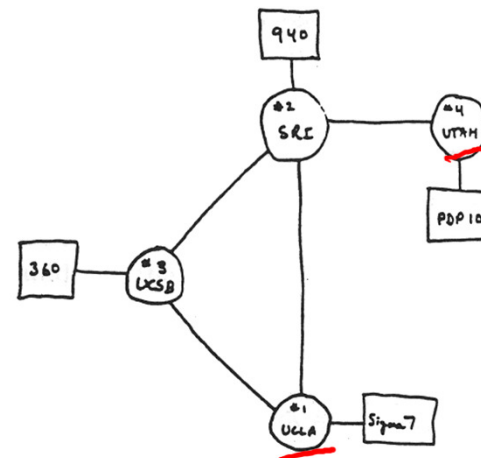
Internet history

1961-1972: Early packet-switching principles

- ❖ **1961:** Kleinrock - queueing theory shows effectiveness of packet-switching
- ❖ **1964:** Baran - packet-switching in military nets
- ❖ **1967:** ARPAnet conceived by Advanced Research Projects Agency
- ❖ **1969:** first ARPAnet node operational

❖ **1972:**

- ARPAnet public demo
- NCP (Network Control Protocol) first host-host protocol
- first e-mail program
- ARPAnet has 15 nodes



Internet history

1972-1980: Internetworking, new and proprietary nets

- ❖ **1970:** ALOHAnet satellite network in Hawaii
- ❖ **1974:** Cerf and Kahn - architecture for interconnecting networks
- ❖ **1976:** Ethernet at Xerox PARC
- ❖ **late 70' s:** proprietary architectures: DECnet, SNA, XNA
- ❖ **late 70' s:** switching fixed length packets (ATM precursor)
- ❖ **1979:** ARPAnet has 200 nodes

Cerf and Kahn' s internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best effort service model
- stateless routers
- decentralized control

**define today' s Internet
architecture**

Internet history

1980-1990: new protocols, a proliferation of networks

- ❖ **1983:** deployment of TCP/IP
- ❖ **1982:** smtp e-mail protocol defined
- ❖ **1983:** DNS defined for name-to-IP-address translation
- ❖ **1985:** ftp protocol defined
- ❖ **1988:** TCP congestion control
- ❖ new national networks: Csnet, BITnet, NSFnet, Minitel
- ❖ 100,000 hosts connected to confederation of networks

Internet history

1990, 2000 's: commercialization, the Web, new apps

- ❖ early 1990' s: ARPAnet decommissioned
- ❖ 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- ❖ early 1990s: Web
 - hypertext [Bush 1945, Nelson 1960' s]
 - HTML, HTTP: Berners-Lee
 - 1994: Mosaic, later Netscape
 - late 1990' s: commercialization of the Web

late 1990' s – 2000' s:

- ❖ more killer apps: instant messaging, P2P file sharing
- ❖ network security to forefront
- ❖ est. 50 million host, 100 million+ users
- ❖ backbone links running at Gbps

Internet history

2005-present

- ❖ ~5B devices attached to Internet (2016)
 - smartphones and tablets *IoT*
- ❖ aggressive deployment of broadband access
- ❖ increasing ubiquity of high-speed wireless access
- ❖ emergence of online social networks:
 - Facebook: ~ one billion users
- ❖ service providers (Google, Microsoft) create their own networks
 - bypass Internet, providing “instantaneous” access to search, video content, email, etc.
- ❖ e-commerce, universities, enterprises running their services in “cloud” (e.g., Amazon EC2) *AI cloud computing*
 - Quantum
 - 6G (2030)

Introduction: summary

covered a “ton” of material!

- ❖ Internet overview
- ❖ what's a protocol?
- ❖ network edge, core, access network
 - packet-switching versus circuit-switching
 - Internet structure
- ❖ performance: loss, delay, throughput
- ❖ layering, service models
- ❖ security
- ❖ history

you now have:

- ❖ context, overview, “feel” of networking
- ❖ more depth, detail *to follow!*